

Math Sprout — Consolidated Specs, as of Sep 25, 2026

Product · Architecture (incl. ML approach) · Design · Kids Trust advice · Learning
Science advice

Prepared for Samarth Sharma · Compiled from the Math Sprout stage-artifact pages in Notion on Fri Sep
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How to read this document. Every Math Sprout stage artifact follows a “supersede, don’t erase” rule: newer decisions are added as dated sections or notes on the same Notion page, and older text stays visible. Each section below reproduces the **current** Notion page in full (formatting lightly cleaned; headings shifted down one level; Notion mentions turned into links). Where a page still contains older text that a later passage replaces, a short grey *compiler’s currency note* at the top of the section tells you which passage wins. Those notes are the only text added by the compiler. All times are Pacific Time (PT).

1. Product Specs

Opportunity memo v1 — Math Sprout (ADOPTED)

Source: Opportunity memo v1 — Math Sprout (ADOPTED)

Status: **ADOPTED v1** (Samarth, 2026-09-21) + additive amendments §11b, §11c, §11d (+ amendments 1 & 2), §11e, §11f; §12 boarded v1.5; §13 PARKED; §14 EXPLORING (not boarded)

Owner / author: Opportunity (PM seat)

Last edited in Notion: Sep 24, 2026, 8:45 PM PT

Compiler's currency note (not part of the Notion source): This memo supersedes DRAFT v0 (not included). Inside the page: §8.1's original "fractions out" wording is superseded by the amendment to grades 2–4 number sense/ops **and** fractions. The Slice 5/7 "home line shows no count" wording is superseded by the §11d amendment (2026-09-24), which allows kid-facing fuel counts. Option C (end-of-session card) is replaced by §11f, and §11d amendment 2 locks option B, the slim fuel strip on the practice screen. The slice scopes you asked about are **§11e** (question variety: templates, 7-day no-repeat window, difficulty steps), **§11f** (session summary, FRAMED) and **§11d amendment 2** (progress/fuel strip). §14 (parent worksheet templates) is **exploring only**.

Status: **ADOPTED (v1)** · Product: Math Sprout (Prod 2) · Owner: Opportunity · Adopted: 2026-09-21 (Samarth) · Gamification locks §9 included · Next: Coherence pressure-test → Interface → Signal · Build plan still gated on Architecture + ML + IA adopted

Audience: Living stage artifact. Supersedes DRAFT v0.

0. Recommendation (one screen)

Build Math Sprout as a motivation + mastery loop for kids practicing math, not as a content library with points bolted on.

Primary user: the child (grade ~2+), practicing alone or with light parent presence.

Secondary user: the parent (trust, consent, progress they can explain in one breath).

Explicit non-primary in v1: teachers / schools / classrooms.

Core thesis: Gamification is the *fuel*; adaptive progression is the *steering*. Neither works alone. Points without honest difficulty become circus. Adaptive difficulty without delight becomes worksheet software kids abandon.

v1 success: a child returns across multiple days, completes focused practice sessions, and shows measurable skill movement on a narrow concept lane — while a parent can see *what* improved without becoming the tutor.

Build approach: concept graph + attempt loop first; progression engine second; gamification economy third (tied to real mastery, not time-on-app); parent trust surface in parallel from day one for kids products.

1. Problem / today → better future

Today

Kids who need math practice usually get one of three broken experiences:

1. **Worksheet / homework grind** — correct/incorrect with little sense of growth; effort feels invisible.
2. **Entertainment apps** — high dopamine, thin skill grounding; parents can't tell if anything was learned.
3. **Adaptive tutors that feel cold** — difficulty adjusts, but the emotional experience is still “get judged by a machine,” so reluctant kids bounce.

Parents sit in the middle: they want progress they can trust, without becoming unpaid tutors every night. Grade labels (“you’re in grade 4”) poorly match real ability — a child can be strong in multiplication and shaky in fractions on the same night.

Better future

The child opens Math Sprout because **practice feels winnable**. Each session has a clear focus, fair challenge (slightly above comfort), immediate encouraging feedback, and a visible “I got better at *this*” signal. When they level up, the app quietly raises the bar on the right concepts — not a random harder quiz.

The parent glances once and understands: practiced X minutes, worked on Y concept, moved from struggling → getting it (or still stuck — honestly). Trust without surveillance theater.

Gap we own

Most products optimize either **engagement** or **pedagogy**. Math Sprout's opportunity is the joint system: **delight that only fires when real learning conditions are met**.

2. Who it's for

Role	Priority	Job to be done
Child (primary)	P0	Feel capable while practicing math; want to come back tomorrow
Parent / guardian (secondary)	P0 for trust surfaces	Know practice happened and mattered; consent / safety; not become the coach
Teacher / school	Later	Class visibility, assignments, standards mapping — out of v1

Age / stage: Grade 2 and above as a *floor*, but the real segment is **ability-banded**, not grade-banded. A “grade 3” label is a coarse prior; the product's unit of truth is **concept mastery + recent performance**, not birth year.

Who we are *not* for in v1: competitive leaderboard kids who need public ranking; parents seeking full curriculum replacement; school procurement buyers.

3. Success signal(s)

Separate **child truth**, **parent truth**, and **business truth**. Don't collapse them into DAU.

North-star (product)

Guided-practice activation with skill lift — within a defined window (propose: 14 days), the child completes $\geq N$ qualifying sessions on $\geq M$ distinct days, and shows improvement on at least one focus concept (pre→post or early→late attempt quality).

Exact thresholds are a Samarth lock + Signal validation later. Directionally: *return + focus + measurable movement*.

Leading indicators (instrument early)

- Session start → first attempt rate (friction)
- Attempts per session + completion without rage-quit
- Return D1 / D3 / D7 (motivation)
- Focus-concept stickiness (are we actually progressing a lane, or random hopping?)
- “Slightly harder” acceptance rate (did adaptive challenge feel fair?)
- Parent open rate of progress summary (trust)

Lagging / outcome

- Concept mastery transitions (Not yet → Getting it → Got it)
- Reduced parent-reported friction (“fight to practice”)
- Optional later: alignment with external benchmarks — **not** a v1 claim

Anti-metrics (do not optimize)

- Raw questions answered (can reward guessing spam)
- Time-on-app alone (can reward stalling)
- Cosmetic streak without qualifying attempts

4. Core product thesis — gamification + adaptive progression as a *system*

Two engines, one loop

Attempt → Feedback → Reward (only if honest) → Next item (difficulty + concept choice) → Identity (“I’m getting better at fractions”)

Adaptive progression answers: *What should this child practice next, at what difficulty?*

Gamification answers: *Why does the child care enough to do another honest attempt?*

If rewards fire on time-on-app or volume, kids game the system and learning dies. If adaptation is invisible and joyless, kids leave before the model has signal.

Product primitives (v1 system objects)

1. **Concept graph** — nodes are teachable skills (e.g. multi-digit addition with regrouping), edges are prerequisites. Grade bands are *views*, not the graph itself.
2. **Item / question bank** — tagged to concept + difficulty + misconception traps. Quality > quantity early.
3. **Learner state** — per-concept mastery estimate + recent form + confidence/uncertainty. (Signal owns how; Opportunity owns what “mastery” means to the kid.)
4. **Session** — short, focus-bounded practice block with one primary concept (maybe one stretch).
5. **Progression policy** — when to stay, remediate prerequisite, or level-up to “slightly harder.”
6. **Reward economy** — XP / badges / streaks / unlocks **gated on qualifying learning events**, not vanity events.
7. **Parent trust surface** — consent, progress narrative, no creepy raw dump.

What “slightly more challenging” means (product rule)

When the child levels up, do **not** jump grade levels. Prefer:

- same concept, harder item parameters; or
- adjacent concept unlocked by prerequisite mastery; or
- mixed review that protects retention.

Challenge should feel like a boss fight they can win with the skill they just practiced — not a pop quiz from a different universe.

Encouragement as design, not copy

Kid-facing feedback is: (a) what went well, (b) one focus, (c) one try-next, (d) progress chip tied to the concept — never shame, never peer comparison in v1.

5. Scope cuts / non-goals

v1 in (sharp)

- Child practice loop + adaptive next-item on a **narrow concept slice** (recommend: one grade-span vertical, e.g. upper-elementary number sense / operations — Samarth locks the slice)

- Parent account, consent, simple progress view
- Gamification tied to mastery events (streaks, XP, concept badges)
- Offline-tolerant or poor-connectivity graceful path for practice continuity (exact tech = Coherence/Build later)

v1 out (explicit)

- Teacher / classroom / LMS / rostering
- Full K–12 curriculum coverage
- Live human tutoring marketplace
- Social / public leaderboards / PvP
- Generative “AI tutor chat” as the primary pedagogy (high risk; can revisit as assist, not spine)
- Parent becoming in-app homework coach with answer keys
- Multi-subject expansion (reading, science)

Later bets (parked, not forgotten)

- Teacher packs / school
- Deeper misconception diagnosis
- Voice / multimodal inputs
- Curriculum standards export for districts
- **Parent agency v1.5 (Samarth · 2026-09-21):** parent **concept steer** first — parent can optionally set tonight’s focus to concepts already in the v1 graph (grades 2–4 number sense/ops **and fractions**). Child still runs Recommended/Challenge/Review via the normal QualifyingEvent loop. **Worksheet OCR later** as convenience on top of steer — maps sheet → existing concept tags only. Explicitly **not**: reproduce the worksheet, answer keys, or any mint path that bypasses QualifyingEvent rules. Out of v1 spine; does not reopen Slice 1–2.

6. Key product bets and risks

Bets (must be true)

1. **Kids will return for mastery-tied rewards** if challenge feels fair — not only for cosmetics.
2. **Ability-banding beats grade-banding** for engagement and learning.
3. **Narrow concept depth beats broad catalog** for v1 learning proof.
4. **Parents need a one-breath progress story** or distribution/trust dies even if kids like it.

5. We can define “qualifying attempt” well enough that gaming the economy is hard.

Risks

Risk	Why it hurts	Mitigation direction
Gamification theater	High DAU, no learning → parent churn / brand damage	Gate rewards on mastery events; anti-metrics
Cold adaptive engine	Correct difficulty, abandoned product	Session UX + encouragement first-class
Content thinness	Adaptation with bad items = noise	Start narrow; item quality bar
Grade-label confusion	Parents buy “grade 4” expecting worksheet parity	Market ability progression; explain clearly
Over-scoping curriculum	Never ships a sharp loop	Hard slice lock
AI-tutor distraction	Shiny chat ≠ practice system	Keep chat out of v1 spine

What must be true before scaling content

- Progression policy produces fair “slightly harder” moves in user tests
- Kids can explain what they’re getting better at
- Parents trust the progress narrative
- Economy isn’t farmable by rapid guessing

7. Recommended sequencing / first-principles build approach

Do **not** start with a full gamification skin or a full curriculum. Sequence for learning about the joint system:

Phase A — Learning loop skeleton (prove the practice unit)

- Concept slice + small high-quality item bank
- Attempt → immediate feedback → next item (manual or simple rules OK)
- Session framing (focus concept, encouraging tone)
- Instrumentation for attempts, correctness, latency, quit points

Exit: kids can complete a session without confusion; we see attempt quality data.

Phase B — Adaptive progression (prove steering)

- Learner state per concept
- Policy: stay / remediate / level-up slightly
- Explicit “you leveled up — here’s the new challenge” moment

Exit: “slightly harder” feels fair in dogfood; focus-lane lift visible.

Phase C — Mastery-tied gamification (prove fuel)

- XP/badges/streaks only on qualifying events
- Identity surfaces (“Fractions explorer”) tied to real mastery
- Anti-farm rules

Exit: return rates rise *without* correctness collapsing.

Phase D — Parent trust (prove keepable product)

- Consent + progress narrative + pause/revoke
- No teacher features

Exit: parents understand progress in one breath; consent path clean.

Parallel craft rule

Interface owns session delight and feedback presentation; Signal owns whether state/progression is ML-shaped vs rules-first; Coherence owns system boundaries. **Opportunity locks the bets above before those loops start** — this memo is that lock request.

Rules-first bias for v1 progression: start with transparent rules + strong telemetry; invite ML when Signal shows rules hit a ceiling (cold start, complex prerequisite inference). Don’t bet the company on a model before the loop exists.

8. Open questions only Samarth can lock

1. ~~Concept slice for v1~~ — **LOCKED 2026-09-21; AMENDED 2026-09-21 (Samarth):** grades 2–4 **number sense / operations AND fractions** (both in fold). Not fractions-only; not ops-only. Prior “fractions out of spine” is **superseded**.
2. **Primary distribution:** direct-to-parent consumer vs education/pilot later — confirm consumer-first?
3. **Positioning:** practice companion vs curriculum replacement (recommend: **practice companion**)?
4. **AI tutor chat:** confirm **out of v1 spine**?
5. **Success thresholds:** approve directional north-star (14-day return + skill lift) and let Signal propose numbers?
6. **Monetization posture for v1:** free dogfood / waitlist vs paid from day one? (affects parent surface urgency)
7. **Brand tone lock:** playful-encouraging vs school-serious (recommend: playful-encouraging, never chaotic)?

8. ~~Companion / BuildGoal metaphor~~ — **LOCKED 2026-09-21**: avatar / warrior power-up (one active BuildGoal). Not pet house.
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9. Gamification locks (Samarth · 2026-09-21)

Status: **Opportunity-owned locks** — refine Interface later; do not reopen without Samarth steer.

Locked

1. **Streak = waning flame (forgiveness built in)**. Missed days cool the flame toward ember; they do not snuff it to zero on day one. Returning to practice rekindles. Goal: pull the kid back, not punish absence. Copy/UX must read warm recovery (“ember still here”), not streak-shame.
2. **Never take XP away in v1**. Progress only slows or pauses; no clawbacks.
3. **Companion mascot + goal-directed build economy**. Include a companion. Loot is **not** random collectibles — pieces assemble toward a goal (e.g. build a pet house, power up a warrior/avatar). Unlocks gate on **qualifying mastery events**, same as XP/badges.
4. **Difficulty is visible**. Show difficulty level. Sometimes offer the kid a choice between difficulty options to invite challenge — with guardrails so easy-mode farming cannot replace adaptive steering (see below).

Opportunity sharpen / pushbacks (adopt unless Samarth overrides)

Streak flame — aligned, with one guardrail. Waning flame is right. Dualingo-style “you broke it” anxiety is the failure mode. Spec: Hot → Warm → Ember across missed qualifying days; Ember persists for a forgiveness window; only long absence goes dormant (not “dead”). Rekindle celebration on return day. Parent surface never frames missed days as failure.

Companion + build goal — LOCKED (Samarth 2026-09-21): v1 metaphor = **avatar / warrior power-up** (not pet house). One active BuildGoal at a time. Each piece unlocks on concept badge / level-up / qualifying streak milestones — never on raw question volume. No paid skips for build pieces in v1. Companion reacts to mastery events; does not nag. Practice remains home; build is a visible side effect.

Difficulty choice — aligned, with anti-farm rules. Default path = **Recommended** (system pick). At defined moments (level-up, stable Got-it, optional challenge day), offer:

- **Recommended** (default)
- **Challenge** (opt-in harder — celebrate harder)
- **Review** (easier / mixed review — limited; does not mint full level-up XP forever)

Kid can choose Challenge to lean in; Review exists for confidence rebuild, not as a permanent easy farm. “Sometimes” ≠ every item; choice appears at session or set boundaries, not per-question spam.

10. Distribution lock (Samarth · 2026-09-21)

v1 ships as a web app first — hostable quickly; shortest path to a working version. Native mobile is out of v1 shape (responsive web OK).

Implications owned elsewhere: Coherence = web client + hosted backend; Build = fast hostable web shipping path; Interface = web-first journeys; Signal = browser latency/inference UX assumptions.

11. Squad refine stamps (Opportunity · 2026-09-21)

Product stamps on peer proposals (do not overwrite Architecture/ML/IA ownership):

1. **QualifyingEvent** is the only celebration/mint contract Opportunity will recognize for XP, streak heat, badges, and build pieces.
2. **Review dual-cap (product intent)**: Review may earn reduced XP only; cannot mint LevelUpSlight, full Badge, or BuildPiece. Got-it / LevelUpSlight require Recommended or Challenge evidence — easy-lane farming cannot fake mastery.
3. **Kid mastery labels (locked with Interface)**: Still learning / Getting it / Got it — no percentages.
4. **Feedback 4-beat (locked with Interface)**: well → one focus → try next → concept chip only if confidence allows. Never “the AI thinks...”.
5. **Practice is home**; companion / build / streak are side effects made visible — not competing home screens.
6. **Streak heat** is QualifyingPracticeDay-only (ember forgiveness per §9).

BuildGoal skin: avatar / warrior (Samarth lock) — one goal.

11b. Consent revoke / pause → pending queue (Opportunity · 2026-09-23)

Additive lock from Beat 15 Frontier SA + Kids Trust + Coherence §22–§23 — **no memo reopen**; thesis holds.

1. **Revoke → drop pending**. Stop new practice immediately; drop offline-queued attempts; **never sync** learning events after consent is gone. Prefer drop over anonymize-then-sync. **Reject** “finish / keep practicing after revoke until sync.”
2. **Pause → hold + parent-visible + quiet resume (LOCKED pick)**. Stop new practice; **hold** flush; parent must see waiting-on-this-device state; on resume, reconcile quietly (credit attempts; **no** celebration dump). Silent hold forbidden. Drop-on-pause is *not* the default.
3. Parent copy: pause/revoke/stop-new ≠ full delete while delete hooks remain deferred — honest now-vs-later.
4. Offline queue = child data at rest: minimize fields (no guardian PII); child-session isolation; purge on sync / logout / profile switch; short TTL.
5. Soft-state: provisional correct/incorrect OK offline; band / Got-it / full celebration = **server-committed only** (fail-closed).

6. Stub thresholds are not kid-facing promises until dogfood locks numbers.

7. **Legal (Samarth):** COPPA/consent disclosure of local pending attempts + accurate pause/revoke effects — **before Slice 2 offline queue ships**.

Architecture/Build: Coherence §22–§23; Slice 5 kid-reachable QE mint gated on replay return shape, celebration fail-closed, and this disposition.

11c. Learning-loop honesty / Slice 2 (Opportunity · 2026-09-23)

Additive lock from Beat 15b Learning Science advisor note — **no memo reopen**; thesis + §11b hold. Soft-state fail-closed, quiet resume, revoke→drop, and no-% ClientView are already locked (§11 / §11b).

1. **oneFocus is diagnostic, not vanity.** Beat 2 of the 4-beat must be **item- or misconception-specific** from item/misconception metadata (deterministic tags). Generic cheer + celebration flash is engagement theater and does **not** earn a Slice 2 learning claim. Interface / Signal stamp sourcing; required before Slice 2 ships the 4-beat as pedagogy.
2. **Offline next-item: cap or freeze.** Short offline queue **or** freeze next-item on the last server-known ability band until sync. Fluency drilling on a frozen pack is OK for Slice 2; claiming mastery / band movement before server commit is **not** (reinforces §11b.5). Full offline adaptive next-item deferred.
3. **Fluency ≠ transfer (claim language).** Slice 2 attempt loop + integrity gates prove **fluency**. Do not market kid/parent “Got it” as transfer (novel format / adjacent application) until item-format diversity + interleaved Review exist (Phase B+). Defer marketing language — **not** a Slice 2 code block if no estimator yet.

Does **not** reopen Architecture spine or block Slice 5 beyond existing celebration fail-closed / revoke disposition gates.

Source: [Learning Science — Slice 2 feedback](#)

11d. Dogfood presentation locks (Opportunity · 2026-09-23)

Additive from Samarth dogfood + squad + Interface stamp — **no thesis / QualifyingEvent reopen**. Soft bands / no-% / Practice-is-home hold.

Feedback frames

1. **Verdict first.** Kid-visible Correct / Not yet chrome before diagnostic beats (renders existing correct). Interface owns language; Kids Trust + Learning Science advise wrong-path tone (calm, not shame).
2. **Distinct frames by outcome (labels differ).**
 - **Correct:** Nice move · Why it works · Try next · Answer — **omit** oneFocus (never “One focus” on correct).
 - **Not yet:** What you tried · One focus (diagnostic) · Try next · Lock in (shows the number).

3. **Anti-theater metadata.** Correct **Why it works** and wrong **oneFocus** only when Item Bank has real solidify / misconception metadata; empty → **omit the beat** (no vanity / no LLM filler). Drop empty correct Lock-in string (“That is the answer we were looking for”).

Fuel — unmissable glance (Samarth lock)

1. **In scope** (§9 / Slice 7) — not deferred. Presentation path = **unmissable glance, not** a fuel tab / second destination.
2. **Home strip** under primary Practice CTA (same column): flame · XP · BuildGoal piece progress — one horizontal strip; Practice CTA stays loudest; strip non-navigating (no fuel tab / shop / companion deep-link).
3. **QE toast fail-closed** — only after server-committed QualifyingEvent mint; short one-breath copy; no toast on volume / Review-only / failed mint.
4. **Polish:** strip visible in pre-QE resting state (don’t hide until earned); one pulse on home return after QE mint. Parent one-breath unchanged — no fuel vanity dump.

Owners: Interface chrome · Build mint-verify then dogfood-hold PR · Signal no ML reopen · Coherence no topology reopen.

12. Parent agency bet — v1.5 (Samarth · 2026-09-21)

Boarded, not in v1 spine.

1. **v1.5:** Parent may optionally **steer** the next session toward concepts already in the locked graph (2–4 number sense/ops **and fractions**). Steering sets focus intent; it does not author items or mint rewards.
 2. **Later:** Worksheet photo/OCR as a convenience that proposes concept tags for parent confirm — never “do this exact sheet.”
 3. **Hard non-goals (any horizon unless Samarth reopens):** answer keys; parent-as-tutor chrome; QualifyingEvent bypass; content outside the active graph neighborhood.
 4. **v1 parent surface stays:** consent + one-breath progress narrative only.
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13. Parked — parent-driven graph growth (v2+) · Samarth aligned 2026-09-21

Status: PARKED — not boarded as an active bet.

Out-of-graph worksheets (puzzles, visual data reasoning, etc.) + LLM-grown categories stay **v2+** only.

Signal pressure-test (2026-09-21): rules + LLM-assist for **proposals only**; promote-to-graph never automatic; sandbox mints = zero QualifyingEvents/XP/badges/BuildPieces; human/editor gate + independent answer check + dup/junk eval before promote; estimator only on stable promoted IDs. Kill silent auto-taxonomy, LLM-only answer keys, sandbox→reward leakage, growth before 2–4 ops has depth.

Do **not** conflate with §12 v1.5 (steer-within-graph). Revisit only after practice spine is proven — or kill if it threatens QualifyingEvent honesty / content thinness.

Decision

ADOPTED by Samarth 2026-09-21 (broader memo + §9 gamification + §10 web-first + concept slice + avatar BuildGoal).

Concept slice LOCKED (amended 2026-09-21): grades 2–4 number sense / operations **AND fractions** (both in fold). Prior “fractions out” superseded.

BuildGoal skin LOCKED: avatar / warrior power-up (one goal).

Parent worksheet ingest: boarded as **§12 v1.5** (concept steer first, OCR later) — out of v1.

Parent-driven LLM graph growth: §13 PARKED v2+ (not boarded) — Signal promote gates apply.

Still open (§8): consumer/monetization/tone/positioning/AI-chat/success thresholds.

Squad: Architecture + ML + IA adopted; Build plan / Slice 1 per Facilitator+Craft gate. No v1 reopen from §12.

ML approach note v0 — Math Sprout (ADOPTED)

IA / lo-fi v0 — Math Sprout (ADOPTED)

Build plan v1 — Math Sprout (practice + mastery loop)

§11d amendment (2026-09-24): kid-facing fuel counts

This supersedes the Slice 5/7 wording “home line shows no count”. That line stays in the record as superseded, not erased.

The child home fuel strip may show numbers, for example  3-day flame ·  120 XP ·  2/5. Samarth asked for visible streak, XP and progress in the 2026-09-23 dogfood, then locked fuel as an unmissable glance.

This fits the anti-volume thesis only when every number comes from qualifying practice. Guards:

- Values are derived read-only from the event bus. The flame counts QualifyingPracticeDay, XP is the sum of QualifyingEvent-pointed credits, and pieces come from the one BuildGoal projection.
- There is no per-attempt XP ticker and no counting of sessions or raw attempts.
- XP never goes down (no clawback, per §9).
- There is no leaderboard or comparison with other children.
- Parent one-breath stays XP-free.

Owner: Opportunity. Coherence stamps the payload field and Signal stamps instrument readability.

§11e Question variety is a dogfood blocker (2026-09-24, Samarth lock)

The gap today. The item bank holds 11 fixed items, one for each skill. The numbers never change, there's no tracking of what a child has already seen, and the items simply cycle. A child sees repeats within minutes, and progression has no harder item to step up to. Growing the bank never appeared on any stage artifact, which was a process miss.

Where we want to be. A grade 2–4 child practicing over about 3 days never feels "I've done this one" or "this never gets harder."

Minimum bar before the dogfood exit test counts:

- Each of the 11 skills has at least 3 templates that ask the question in genuinely different shapes, not the same problem with new numbers. Operands are fresh on every draw, and the code computes the answer (no LLM).
- Each skill offers at least 3 difficulty steps across its templates, so the band logic can move a child up at least twice past the entry level.
- No child sees an identical item (same template and same operands) twice within 7 days, and the same template never appears back-to-back within a session.
- Every template carries a misconception tag and a oneFocus. The §11d anti-theater rule still applies: if whyItWorks metadata isn't real, that beat is omitted.

Out of scope for this cut: LLM-generated items, word problems, and visual fraction models. Prompts stay numeric and symbolic. This is not Slice 8.

Success signal: zero exact repeats across the real-kid test, plus at least one real difficulty step up in each skill the child practiced.

Dogfood exit bar amended: this variety bar has to be met before the 3-day real-kid test counts toward exiting the hold.

Owners: Opportunity owns the bar. Signal owns the taxonomy and the difficulty scale, Coherence owns where the generator runs and how items are identified, and Build owns the plan change for Craft.

§14 EXPLORING, not boarded (2026-09-24): parent-authored templates from worksheets

This refines the §12 line "worksheet OCR later". §13 (new categories grown by AI) stays PARKED for v2+.

Who it's for. A parent of a grade 2–4 child who wants this week's practice to match what the class is doing, inside the app the child already likes.

Today. Parent steer (§12, v1.5) can point practice at an existing skill, but the questions come only from our templates. A school worksheet may use a problem shape we don't have.

Better future. The parent snaps a worksheet, and within minutes the child is practicing the same kind of problem with fresh numbers. It is always correct and never a copy of the worksheet or its answers.

Options:

1. Steer only (§12 as boarded). The parent picks the skill, and no worksheet is involved.
2. Match by example. AI reads the photo and maps it to an existing skill, step and template. Nothing new is authored. This is §12's later worksheet reading.
3. Author inside the graph. AI proposes a new template for an existing skill in the §11 format. Code validates it by generating samples and computing the answers, and the parent previews and approves it. The child's path uses no AI.
4. New skills or categories. This is §13 and stays parked.

Recommendation. Keep 1 as boarded. Explore 2 and then 3 after the dogfood hold exits and PR A's template format lands. This is not Slice 8 and not a blocker. Before building option 3, run a cheap evidence check on about 20 real worksheets: what share maps to an existing template (option 2 is enough), what share needs a new pattern inside a skill (option 3 is justified), and what share falls outside the graph (§13).

Guards if option 3 is built:

- A parent-added template never feeds Got-it or band evidence until Signal promotes it.
- Until promoted, it earns reduced XP and no BuildGoal pieces, like Review, so a parent can't create easy fuel by accident.
- Every sample answer is computed by code. A template whose samples fail validation can't be approved.
- Photos are processed and discarded, never stored, and names of the child, teacher and school are never kept.
- Worksheet answer keys are never reproduced.

Success signal. The child reaches qualifying practice on a parent-added template in the same week it was added, zero templates with a wrong answer ever reach the child, and parents approve most proposed templates without editing them.

Open, for the seats: Signal owns the promotion rule and guarding against a wrong skill or step. Coherence owns the authoring boundary (AI only at authoring time, server-side validation). Kids Trust owns photo handling.

§14 guards added from Kids Trust (2026-09-24). Opportunity stamped these; Kids Trust advises.

Photos are child data as a whole, including handwritten answers and anything in the background.

- We keep only approved templates. We never keep the image, the OCR text, or the child's answers.
- Deletion happens as soon as the proposal comes back, with a retry window measured in hours.
- The vendor must not train on the images and must keep them only minimally. EXIF and location data are stripped.

- Before the first upload, the parent sees a one-line notice and a nudge to crop the photo to just the questions.
- There's no new purpose. School answers never set band and never build a profile.

Templates capture structure and skill only. Code rejects any template that reproduces the source item. Answer keys are never ingested, and no publisher or school branding is kept.

The child sees only generated items, under the same ClientView, verdict, fail-closed and no-shame rules. The source is never named (no "from your worksheet" and no "your teacher"). Parent approval can't bypass validation or banding, and items enter normal progression; the earlier guard that they don't count as Got-it evidence until Signal promotes them still stands. Any wording a parent can edit goes through the never-list.

The ~20-worksheet evidence check follows the same photo rules: photos are processed, then discarded.

Nice-to-have: on-device crop or redaction, and no sharing of templates beyond the household.

Blocks wider distribution only, parked under the personal-software rule: COPPA notice and consent, a vendor data agreement for third-party image processing, teacher and school names never retained, and a copyright policy that allows derived templates only.

§14 converged with peers (2026-09-24). Still EXPLORING, not boarded.

- **Coherence (Architecture §28):** AI writes only a declarative template spec, never code and never answers. The server validates it on samples before the parent sees it. Template versions never change once saved, and `evidence_eligible` is copied onto each issued item.
- **Build:** PR A's template format is that same spec, plus `version`, `provenance` and `evidence_eligible` columns from day one, so parent templates later are new rows, not a second format.
- **Signal (ML note, exploring):** A parent template starts as Review-only and not evidence-eligible. It is promoted after at least 20 attempts over at least 3 days, accuracy within about 15 points of our own templates at the same skill and step, and at least a third of wrong answers matching its declared bug rules. It gets demoted back to Review if it drifts.
- **Interface:** A parent approval screen shows three sample problems with answers, plus the skill and step in plain words. The child sees no badge.
- **Opportunity exposure rule (new):** While a template is in shadow, a parent steer toward that skill guarantees it a share of Review slots that week, within the existing Review caps. Otherwise the child rarely sees it and it never reaches promotion.
- **Success signal revised:** The child practices a parent-added template in the same week it was added (qualifying practice isn't required, since Review-only can't light the flame by itself). Zero wrong-answer templates reach the child, and most proposals are approved without edits.
- **Sequencing unchanged:** after the dogfood hold exits and PR A lands, gated on the ~20-worksheet evidence check.
- **Exposure rule limits (Signal, accepted by Opportunity):** The guaranteed share is capped at about a third of Review slots, so a steer never pushes out the child's due reviews on weak skills. Promotion

usually lands in week two or three. The same-week practice signal is separate from the promotion gate, and the gate is not shortened to fit the week.

§14 difficulty calibration (2026-09-24, Samarth input; still EXPLORING)

- **At upload**, the parent is asked one question per worksheet: its grade level. This is the first calibration and also a scope check. A worksheet outside grades 2–4, or one that maps to no skill in the graph, is declined as out of scope (it belongs to §13, which is parked) rather than stretched to fit.
- **Per templated question**, the parent confirms difficulty in plain words (Easy, Medium or Hard, where multi-step means Hard). To keep the parent's effort low, it is pre-filled from the step features code computes, and the parent confirms or overrides.
- **Reconciling**. The parent's grade and difficulty are a prior, checked against code-checkable step features (number of steps, regroup, unlike denominators, blank position). When they agree, the step is assigned. When they disagree, the approval screen flags it and the parent picks, and the code-computed step wins if the parent skips.
- **Logging**. The template, step and parent prior are logged on every attempt from day one.
- **Later, at scale**: statistical calibration from correct-answer distributions across many learners, using an ability-adjusted model. With one child, correct rates mix up ability with item difficulty, so single-kid data can flag a template as possibly mis-rated but never re-rates it automatically. This is consistent with Signal's lock of no IRT until there's data.

§11d amendment 2 — F1 option B LOCKED (Samarth, 2026-09-24)

Samarth locked option B: a slim fuel strip on the practice screen plus a reward line after each correct answer that mints fuel. The hi-fi set is treated as signed off. Color principle: flame heat colors (hot #E4572E, warm #F29E4C, ember #B5651D, resting #9AA0A6), XP gold #E8B923, pieces violet #7C5CBF, green #2E9E5B only for Correct, Not yet blue #4A7FC1, neutral slate step badges, ink #1F2A44 buttons. Flame heat depends only on how recent the last qualifying day was: hot means today, warm means yesterday, ember means one missed day (the count holds), and resting means two or more missed days (it reads "Flame resting", never 0). Option C (end-of-session card) is replaced by §11f.

§11f — Session summary (FRAMED, not built)

This is Samarth's ask: a gym-app-style summary at the end of each practice session.

- **Placement**: fits inside the dogfood hold because it only presents existing data. It mints nothing, changes no progression, and is not Slice 8. It queues after #10, #12 and the option B follow-up, and needs #12 before it can show new question types. It doesn't count toward the exit bar and can't delay PR A or PR B. Any stat that needs a new data path is cut from v1.
- **What "new record" means**: the child got better, not that they did more. Allowed records: first correct answer at a higher step on a skill; first Got-it on a skill; first time seeing a new question type (after #12); longest correct run in a session, counting only Recommended and Challenge items; new longest flame. Minutes and questions practiced appear only as plain facts in small text, never as records or goals.

- **Guards:** every number comes from qualifying practice only (no unreadable-answer rejects, legacy items or replays; Signal sets the rules). The child is compared only with their own past, and the card never shows a decrease. At most two records; a session with none shows what's next, never "No records today." Same color map as §11d. Read-only, with the same numbers as the practice-screen strip. Parents see no XP.
- **Success signal:** most sessions show at least one record that reflects real improvement, and the share of fast guesses doesn't rise after launch. If it rises, cut the count lines first.
- **Next:** Signal decides which stats can be computed honestly, Coherence flags any new data path, and Interface draws the lo-fi.

2. Architecture Specs

2A. Architecture note v0 — Math Sprout (practice + mastery loop)

Source: [Architecture note v0 — Math Sprout \(practice + mastery loop\)](#)

Status: **ADOPTED v0** (2026-09-21) + supersessions / fold-ins §17–§31a (latest: §31a addendum 3, 2026-09-25)

Owner / author: Coherence (SA seat)

Last edited in Notion: Sep 25, 2026, 1:46 AM PT

Compiler's currency note (not part of the Notion source): The header lines “Does not unlock Build plan” and §15 “Status remains DRAFT” date from the first pass. §16 then set the status to ADOPTED (v0). §17 (online-only) is superseded by §18 (offline queue restored). §19 (fractions out) is superseded by §20 (number sense/ops **and** fractions). §22's strict server-time rule is amended by §27 (offline day attribution). §31's step-up fallback is superseded by **§31a**, which switches skill silently and falls back to a logged repeat with no step-up. §28's parent-authoring parts are exploring only. A separate duplicate page, “SUPERSEDED — Architecture note v0 (duplicate; use canonical)”, is not included.

Status: **ADOPTED (v0)** · Product: Math Sprout (Prod 2) · Owner: Coherence · Adopted: 2026-09-21
(Opportunity stamp + Samarth web-first lock) · Source: [Opportunity memo v1 — Math Sprout \(ADOPTED\)](#)

Does not unlock Build plan. Interface / Signal still Facilitator-gated after this first pass.

Gamification §9 locks are product locks — feasibility challenged below; not overwritten.

1. Intent (system shape)

Math Sprout is a **motivation + mastery loop** for kids practicing math: concept-graph practice sessions with adaptive next-item steering, a mastery-gated reward economy, one companion/build-goal as a first-class product system (not chrome), and a parent trust surface from day one.

In v1: child practice loop on a narrow concept slice → learner state + progression policy → qualifying-event rewards (+ streak ember + companion build) → parent consent + one-breath progress narrative.

Out of v1 core: teachers/schools/LMS, full K–12 catalog, social/PvP leaderboards, generative AI tutor chat as spine, parent-as-coach answer keys, multi-subject.

Primary user = child (ability-banded, grade ~2+ as floor). Account root of trust = parent/guardian (consent + payer). Teachers = later boundary — do not leak child learning telemetry into shared classroom spaces in v1.

Thesis translation to architecture: two engines must share one event bus of **qualifying learning events**. If rewards and progression disagree on what an attempt “counted,” the joint system collapses into either circus or worksheet software.

2. Bounded contexts

Context	Owns	Does not own
Identity & Consent	Guardian account, child profiles, age/COPPA-class consent, pause/revoke/delete/export	Item pedagogy; reward copy
Concept Graph	Nodes (teachable skills), prerequisite edges, grade-band <i>views</i> over the graph	Mastery estimates; session UX
Item Bank	Items tagged concept + difficulty + misconception traps; quality bar	Progression policy
Learner State	Per-concept mastery estimate, recent form, uncertainty/confidence	Reward minting; parent copy
Practice Session	Focus-bounded block, attempt loop, feedback delivery, quit/complete	Long-term curriculum planning
Progression Policy	Stay / remediate prerequisite / slight level-up; when to offer difficulty choice	Cosmetic rewards
Reward Economy	XP, badges, streak ember state machine, companion build unlocks — only on qualifying events	Defining pedagogical mastery bands (Opportunity + Signal)
Companion / Build Goal	One active metaphor + goal assembly progress; reactions to mastery events	Random loot tables; paid skips
Parent Trust	Consent UX data, progress narrative aggregates, no creepy raw dump	Child session delight
Telemetry / Anti-farm	Attempt integrity signals, farm detectors, qualifying-event definitions as executable rules	Product north-star thresholds (Samarth + Signal)

Teacher/classroom is a **future** bounded context. Do not model rostering or assignment as flags on v1 child sessions.

3. Core runtime loop (system object graph)

```
GuardianConsent → ChildProfile
  → OpenSession(focusConcept)
  → SelectItem(progressionPolicy, learnerState, optionalDifficultyChoice)
  → Attempt → Score/Feedback
  → UpdateLearnerState
  → Emit QualifyingEvent? → RewardEconomy + CompanionBuild + StreakEmber
  → NextItem or SessionComplete
  → ParentTrust aggregates (derived only)
```

Hard coherence rule: `QualifyingEvent` is a first-class schema shared by Progression, Reward, Companion, Streak, and Telemetry. No parallel “soft XP” path that bypasses it.

4. What must be true architecturally

1. **Single source of truth for concept identity** — graph node IDs stable; grade labels are projections, never primary keys for learner state.

2. **Learner state is per-child × per-concept** — not per grade, not global ELO alone.
 3. **Sessions are focus-bounded** — one primary concept (+ optional stretch); hopping without policy is a bug, not a feature.
 4. **Progression policy is explicit and explainable** — rules-first v1 (Opportunity bias); Signal may later propose ML only when telemetry shows ceiling.
 5. **Difficulty is a tagged item property + policy decision** — visible to the child; choice moments are session/set boundary events, not per-item spam.
 6. **Reward minting is idempotent and server-authoritative** — client can animate; client cannot invent XP/pieces.
 7. **Companion/build is product state** — persisted goal progress; not a skin theme. One active build goal; pieces map to qualifying milestones.
 8. **Streak is a state machine (Hot → Warm → Ember → Dormant)** — not a binary counter. Ember has a forgiveness window; parent surface never frames miss-as-failure.
 9. **Parent trust consumes derived narratives** — practiced minutes, focus concept, mastery band movement — never raw attempt dumps or “got question 47 wrong.”
 10. **Offline / poor connectivity** — attempts can queue locally; reward/progression reconciliation is server-side on sync (no double-mint).
 11. **Anti-farm is architectural** — qualifying definitions + rate limits + Review-lane XP caps live next to the economy, not as Interface copy.
-

5. v1 boundaries (sharp)

In

- Narrow concept slice (Samarth locks neighborhood) + small high-quality item bank
- Attempt → feedback → next item with rules-first progression
- Mastery-gated XP / concept badges / streak ember / one companion build metaphor
- Difficulty visible; Recommended / Challenge / Review choice at **defined** moments only
- Parent account, consent, one-breath progress view, pause/revoke/delete
- Graceful poor-connectivity path for practice continuity

Out

- Teacher/classroom/LMS/rostering
- Full curriculum coverage; multi-subject
- Public leaderboards / PvP / social graph

- Generative AI tutor chat as primary pedagogy
- Parent answer keys / in-app homework coach
- Paid skips for build pieces
- Dual build metaphors (pet house *and* avatar) — pick one later; architecture assumes **one**
- Taking XP away / clawbacks

Non-negotiable no-gos

1. Rewards on raw volume, time-on-app, or non-qualifying guesses
2. Per-question difficulty choice spam that lets easy-mode replace adaptive steering
3. Streak shame / zero-on-miss-day (violates \$9 ember lock)
4. XP clawbacks in v1
5. Companion nagging or monetized piece skips
6. Grade-band as the only progression key
7. Child self-consent for under-13 data collection (COPPA-class)
8. Behavioral ads / selling child learning data
9. Cross-child public rankings that identify minors
10. Quiet “AI tutor chat” spine without reopening Opportunity + this note

6. Gamification locks — feasibility pressure-test (do not overwrite)

Lock	Feasibility read	Architectural implication
Streak = waning flame	Buildable as explicit state machine + calendar of <i>qualifying</i> days (not app-open days).	Persist <code>streak_state</code> , <code>ember_expires_at</code> , <code>last_qualifying_day</code> . Rekindle is an event, not a counter reset. Parent aggregations must use warm copy keys, not “broken streak.”
Never take XP away	Easy if ledger is append-only.	XP ledger = credit-only; corrections use “do not mint” not reverse. Cap Review-lane XP instead of clawback.
Companion + goal-directed build	Coherent if build progress is a projection of qualifying milestones. Risk: twin economies (XP vs pieces) drift.	One <code>BuildGoal</code> aggregate; piece unlock rules reference the same <code>QualifyingEvent</code> taxonomy as XP/badges. No second random loot table.
Difficulty visible + occasional choice	Highest coherence risk (anti-farm).	Choice offered only at <code>SessionBoundary</code> or <code>SetBoundary</code> events. Default = Recommended. Review lane: limited uses + reduced/non-level-up XP. Challenge lane: celebrate; still must meet integrity checks. Policy remains authoritative for next Recommended item even after a Review session.

Challenge (not overwrite): §9 is product-correct; without a shared qualifying-event schema and Review caps, difficulty choice + streak forgiveness become farm vectors. Architecture makes those guards first-class.

7. Qualifying events (v0 taxonomy — Signal/Opportunity refine thresholds)

Executable categories the economy and progression both consume:

1. **HonestAttempt** — submitted answer with integrity ok (not empty/spam/timing anomaly); counts toward session stats.
2. **ConceptProgressTick** — measurable movement on focus concept within session or across early→late attempts (Signal defines estimator).
3. **MasteryBandTransition** — Not yet → Getting it → Got it (Opportunity owns kid-facing meaning).
4. **LevelUpSlight** — policy-approved slight difficulty/concept advance with celebratory boundary.
5. **QualifyingPracticeDay** — ≥ 1 session meeting minimum honest attempts on a calendar day (drives streak heat).
6. **BadgeMilestone / BuildPieceUnlock** — mapped to band transitions, level-ups, or qualifying streak milestones — **never** raw question count.

Anti-events (must not mint full rewards): rapid guessing patterns, Review-lane overuse, abandoned sessions below attempt floor, duplicate sync retries.

Exact numeric thresholds = Samarth lock + Signal validation later. Architecture requires the **types** and mint gates now.

8. Coherence risks

Risk	Why it matters	Mitigation
Difficulty-choice farm	Kid lives in Review; adaptive steering dies; parents see fake progress	Boundary-only choice; Review XP/level-up caps; Recommended remains default; telemetry on choice mix
Streak ember vs qualifying day	If streak heats on app-open, §9 becomes theater	QualifyingPracticeDay only; Ember timer independent of opens
Twin economies	XP says “up,” build pieces lag (or vice versa) → trust break	Single QualifyingEvent bus; build pieces are projections
Concept graph vs grade marketing	Parents buy “grade 4”; state is ability-banded	Graph is source of truth; grade = view; parent narrative speaks concepts
Cold rules progression	Fair difficulty but joyless → churn before Signal has data	Session/Interface delight parallel; don’t delay Phase A for ML
Content thinness	Policy over noisy items = thrashing	Narrow slice + item quality bar before wide graph
Offline double-mint	Local XP then sync race	Server ledger idempotency keys per attempt

Risk	Why it matters	Mitigation
Parent surveillance feel	Detailed wrongness dumps → kid anxiety / parent coach mode	Aggregates + concept story only in v1
Companion as chrome	Skin without state → Opportunity thesis fails	Persist BuildGoal; gate pieces; Craft/Build treat as domain aggregate
Premature AI tutor	Chat bypasses attempt loop	Hard out of spine; reopen memo + this note to add
Later teacher bolt-on	Sharing flags on existing sessions break consent	New context + new consent — not a permission bit

9. Phased architecture (maps to Opportunity A–D)

Phase	System prove	Architectural deliverable
A Learning loop	Practice unit works	Concept slice + item bank + session state machine + attempt instrumentation
B Adaptive steering	Slightly-harder feels fair	Learner state store + progression policy service + level-up boundary event
C Mastery fuel	Return without correctness collapse	QualifyingEvent bus + XP ledger + streak SM + one BuildGoal
D Parent trust	Keepable product	Consent/lifecycle + derived progress narrative APIs

Do not ship C's cosmetics before A's attempt integrity exists. Do not let D wait until “after growth” — consent is on the critical path for kids products.

10. Placement sketch (v0 — not a vendor lock)

Layer	v0 placement	Notes
Child practice client	Mobile/web client	Offline queue for attempts
Session + scoring	Server	Authoritative scoring for banked items; integrity checks
Learner state + progression	Server	Rules engine first; Signal may swap estimators behind interface
Reward + streak + build	Server	Append-only ledger; client renders
Parent trust	Server + parent client	Separate surface; minimization
Item/concept authoring	Ops/CMS later	Start with curated pack; not user-gen

Inference/ML: **not required for Phase A**. Signal owns whether mastery estimation becomes model-shaped; until then rules + telemetry.

11. Implications by role

- **Opportunity:** Thesis holds. Keep §9 locks. Still need Samarth concept-slice + build-metaphor picks (non-blocking for this v0 shape). Do not expand to teacher or AI-chat spine without reopening.

- **Interface:** Own session delight, visible difficulty, ember recovery copy, companion reactions, parent one-breath narrative — within anti-farm boundaries (choice frequency, no streak-shame).
- **Signal:** Propose mastery/form estimators and integrity features **inside** rules-first progression. Challenge when rules hit ceiling. Help define QualifyingEvent thresholds — do not invent a parallel reward signal.
- **Build:** (later, after Architecture + ML + IA adopted) Session SM, ledgers, idempotent sync, consent gates. No Build plan unlock from this note alone.
- **Craft:** When plan/code exists — ledger integrity, anti-farm enforcement, consent/delete paths as security-sensitive.
- **Facilitator:** Fan-out Interface → Signal after Opportunity stamps this note (or challenge list). Hold Build plan gate.

12. Open questions (non-blocking for v0 shape)

1. v1 concept slice neighborhood — **Samarth**
2. Companion metaphor (pet house vs avatar) — **Samarth / Opportunity**
3. Numeric north-star thresholds — **Samarth + Signal**
4. COPPA verification method — **Legal / Samarth**
5. Client delivery (PWA vs native) — **Build later**; keep domain logic delivery-agnostic
6. Whether any on-device scoring matters — likely no for banked math items; revisit only if handwriting/voice input appears (out of v1 spine)

13. Decision log (v0)

Decision	Choice	Alternative rejected
Product spine	Attempt loop + progression + mastery-gated economy	Content library + bolt-on points
Progression	Rules-first, explainable	ML-first before loop exists
Rewards	Shared QualifyingEvent bus	Separate cosmetic counters
Streak	Hot/Warm/Ember/Dormant on qualifying days	Binary break-on-miss
XP	Append-only; no clawbacks	Punitive debit economy
Companion	One BuildGoal domain aggregate	Cosmetic mascot only / dual metaphors
Difficulty choice	Boundary moments + Review caps	Per-question free pick
Parent data	Derived narrative	Raw attempt surveillance
Teacher	Out of v1 context	Early rostering flags

14. Challenges back to Opportunity (stamp or refute)

1. Adopt **QualifyingEvent** as the joint contract between progression and economy — or name the alternative single contract.
2. Confirm Review lane is **capability-limited** (not infinite easy XP) — required for §9 difficulty choice to stay coherent.
3. Confirm streak heat uses **QualifyingPracticeDay**, not opens — required for ember lock honesty.
4. Keep Build plan gated until Architecture (this) + ML + IA adopted — unchanged.
5. Non-blocking: pick build metaphor + concept slice when ready; architecture assumes one metaphor and a narrow graph.

Next: Opportunity stamps or challenges. Facilitator unlocks Interface (journeys) then Signal. Coherence holds shape; reopen only on constraint change or Samarth tradeoff.

15. Stamps / deltas (2026-09-21 room)

From Build: Phase A must ship attempt-level idempotency keys (offline sync). Review “capability-limited” = dual unit — sessions/week ceiling + cannot mint LevelUpSlight or full Badge/BuildPiece (reduced XP only). One BuildGoal; refuse second XP/pieces path.

From Signal (ML v0): Rules + telemetry behind swappable learner-state interface; QualifyingEvent only mint path. Got-it / LevelUpSlight require Recommended or Challenge evidence.

From Interface (IA v0): Practice is home; companion/build/streak are visible side effects. Coherence stamps: **SessionBoundary/SetBoundary** = only difficulty-choice mounts; **parent surface must not deep-link attempt history in v1**.

Status remains DRAFT pending Opportunity stamp on §14 + Interface 4-beat / Still learning·Getting it·Got it labels.

16. Samarth lock — web-first distribution (2026-09-21)

Locked: Math Sprout ships as a **web app first** (hostable, shortest path to working). Architecture assumes **web client + hosted backend**. Native mobile is **out of v1 shape** (may wrap later; domain logic stays delivery-agnostic).

Also folded as adopted constraints: QualifyingEvent-only mints; Review dual-cap; Got-it/LevelUpSlight require Recommended or Challenge evidence; SessionBoundary/SetBoundary-only choice mounts; parent surface no attempt-history deep-links; Phase A attempt idempotency; one BuildGoal as bus projections.

Adoption: Opportunity stamped product intent on these; Architecture note status → **ADOPTED (v0)**. Build plan still gated on ML + IA adoption (both have v0 up — Facilitator owns unlock order).

17. Supersession — connectivity (2026-09-21 Craft Review)

Was: v1 includes offline/poor-connectivity attempt queue with server idempotent sync.

Now (v1): Online-only for attempts — network required to submit. Offline queue deferred to a named later slice (Build §10 / Craft M3).

Tradeoff: Faster Slice 1–2 ship; school Wi-Fi / flaky networks may bounce kids mid-session until queue lands.

Also stamped: SessionBoundary-only for difficulty choice in v1 (SetBoundary deferred) — stricter, still coherent.

Do not claim offline-tolerant in v1 product or Architecture summaries until the queue slice ships.

18. Supersession — connectivity restored + feedback contract (2026-09-21)

§17 online-only is superseded. Build restored Slice 2 **client offline/poor-connectivity attempt queue** + sync reconcile via idempotency (Build plan §10 M3). Architecture continuity path is again: queue locally → server mint/idempotent replay.

Attempt response contract (Architecture + IA + ML): correct + ClientView (bandLabel · showConceptChip · celebrationTier) + **4-beat keys** (whatWentWell · oneFocus · tryNext). Never score % / raw confidence. Concept chip only via showConceptChip.

SessionBoundary-only (SetBoundary deferred) unchanged.

19. Samarth locks — content + skin (2026-09-21)

Concept slice (v1): grades 2–4 **number sense / operations** only. Fractions are **not** the v1 spine. Architecture: Concept Graph seed pack / Item Bank for v1 bound to this neighborhood; grade labels remain views over the graph.

BuildGoal skin (v1): avatar / warrior power-up — one active BuildGoal. Not pet house. Architecture: structure unchanged (pieces = QualifyingEvent projections only; no twin loot table); skin is presentation, not a second economy.

No topology reopen. Stub pack for Slice 2 targets the number-sense/ops neighborhood.

20. Samarth amendment — concept slice (2026-09-21)

Supersedes §19 fractions-out. v1 Concept Graph / Item Bank neighborhood = grades 2–4 **number sense / operations AND fractions** (both in fold). Not fractions-only; not ops-only.

Shape check: No topology change — same Concept Graph + Learner State + QualifyingEvent bus; seed pack widens. No Craft M1–M5 reopen; Slice 1 unchanged; BuildGoal/skin unchanged.

21. Supersession — versioned policy + offline eval (2026-09-22)

Source: Frontier SA advisor note (eval harness / policy_version) — advises only; Coherence stamp below.

Architecture lock:

1. Persist `policy_version` on every scored attempt/session (constant `rules-v0` OK until Phase B). Required before Phase B claims; stub now.
2. **Production path** = live `MasteryEstimator` + progression policy serving kids. **Eval harness** = offline consumer of the attempt log / fixtures — **not** a second live scorer. Twin scoring paths are a coherence failure.
3. Soft-state remains an eval contract too: fixtures assert `ClientView` never leaks % / raw confidence / AI-judgment copy.
4. Integrity fixtures (empty / too-fast / duplicate key / identical spam; Review mint-gate when bus lands) are architectural anti-farm proof, not polish.

Build timing stamp: Slice 2/3 already on `main` — folding `policy_version` + `ClientView`/integrity fixture gap-check into **Slice 4 before QualifyingEvent economy is kid-reachable** is an acceptable supersession of “fixtures exit Slice 2.” Do not mint full XP/Badge/BuildPiece to kids without those fixtures + versioned attempts.

Signal owns harness + baseline dataset location; Build owns instrumentation contract + fixtures. No ML-in-spine reopen.

22. Supersession — Slice 2 reliability + consent lifecycle (2026-09-23)

Source: Frontier SA advisor notes (reliability / stage shape / consent-delete / eval) — advises only; Coherence stamp below. **Disposition: no-reopen** of spine (QualifyingEvent-only mint; soft-state `ClientView`; offline eval ≠ twin scorer; §18 queue; §21 `policy_version`). Clarifying contracts that were implied but under-specified.

Architecture locks:

1. **Same-key replay return shape.** Same `attempt_idempotency_key` returns the **original** attempt result + `ClientView` + already-emitted event ids (no-op mint). Never an empty 409 / “try again” that forces a second attempt id. Broken replay now = double-mint later.
2. **Server-authoritative time.** Server `accepted_at` / reconcile sequence owns session day and later `QualifyingPracticeDay`. Device wall clock must not decide heat. Day boundaries use **child.timezone on the server** (schema stub early).
3. **celebrationTier fail-closed.** Derive only from what this server transaction commits. Fail closed to none / quiet. No full chrome without a committed result (even before XP bus).
4. **Pause / revoke → pending queue. Default:** pause → stop new practice + **hold flush**; revoke → **drop or anonymize** pending payloads — do **not** sync learning events after consent is gone. “Finish the session then stop” is an **Opportunity + Kids Trust** tradeoff, not a silent eng choice.

5. **Offline queue = child data at rest.** Minimize / encrypt fields on device; purge on successful sync; never log raw answers to third-party crash reporters from the queue path. Field allowlist for analytics/crash SDKs before production kids traffic.
6. **Consent lifecycle (appendix, not spine reopen).** Revoke ≠ delete. Revoke immediately stops practice + mint paths. Delete/export requires an enumerated cascade graph (Guardian → ChildProfile → Sessions → Attempts → Events → XP/Build projections → Parent caches → device queues) with cascade | anonymize | retain-with-legal-basis per edge. COPPA verification **method** remains **Samarth + Legal** — Architecture owns hooks, not method theater. Teacher remains a **future context + new consent**, not a flag on child sessions.
7. **§21 unchanged.** `policy_version` stub; eval = offline attempt-log consumer; integrity + ClientView leak fixtures before kid-reachable QualifyingEvent economy.

Slice 5 gate: If Slice 5 opens QualifyingEvent economy to kids, **block kid-reachable mint** until (a) same-key replay fixtures, (b) pause/revoke+queue disposition implemented or explicitly gated, (c) §21 `policy_version` + integrity/soft-state fixtures. Delete/export cascade + COPPA method block **production / trust marketing**, not dogfood engineering if those claims stay off.

Not reopened: QualifyingEvent bus; soft-state; Review dual-cap; Got-it / LevelUpSlight evidence gates; web-first; concept slice; avatar BuildGoal skin; no ML-in-spine.

23. Fold-in — Kids Trust Slice 2 (2026-09-23)

Source: Kids Trust advisor note (consent / offline queue) — advises only. **Disposition: no-reopen.** Tightens §22 pause/revoke + queue contracts; aligns with Frontier SA reliability.

Architecture locks (additive to §22):

1. **Revoke → drop pending** (never sync after consent is gone). Prefer drop over anonymize-then-sync for trust; anonymize-at-rest only if Legal requires retention without learning use.
2. **Pause → hold or drop — pick explicitly (Opportunity).** If hold: **parent-visible** “waiting on this device until resume” + **quiet resume reconcile** (credit attempts; no celebration dump / bulk chrome). **Silent hold is forbidden.**
3. **Queue hygiene on shared devices:** minimized fields (no guardian PII in queue blob); child-session isolation (sibling flush cannot attach attempts to the wrong child); purge on successful sync, logout, and profile switch; short retention / TTL.
4. **Parent narrative honesty (Architecture truth):** pause/revoke/stop-new ≠ full delete while delete hooks remain deferred. Parent Trust / Interface copy must match executable behavior — not imply erase-everything.
5. **Soft-state / celebration:** provisional correct/incorrect + 4-beat template may show offline; band / Got-it / full celebration = **server-committed only** (§22 fail-closed).
6. **Legal gate (Samarth):** COPPA/consent disclosure must cover **local device storage of pending practice attempts** and accurate pause/revoke effects — **before Slice 2 offline queue ships**

(stronger than “before trust marketing”). Full delete cascade remains deferred only if narrative is honest and revoke = drop.

Not reopened: QualifyingEvent spine; soft-state ClientView; §18 queue; §21 eval/policy_version; §22 replay + server time.

Slice 5 / kid-reachable mint gate unchanged from §22. Offline queue ship additionally blocked on Samarth/Legal disclosure lock above.

24. Fold-in — Learning Science Slice 2 (2026-09-23)

Source: Learning Science advisor note (feedback / offline learning-loop) — advises only. **Disposition: §24 additive · no-reopen · no Opportunity debate required** for the contracts below (Opportunity already locked pause = hold). Fluency-vs-transfer mastery marketing remains Opportunity language (defer), not a topology change.

Already held (§18 / §22 / §23): soft-state ClientView (no % / confidence); offline provisional correct/incorrect + 4-beat OK; band / Got-it / full celebrationTier server-committed only; revoke → drop pending; pause = hold with parent-visible waiting + quiet resume reconcile.

Architecture additives:

1. **Offline banding drift guard.** Unbounded offline adaptive next-item is out of Slice 2. Sessions must use **at least one** of: (a) a short **offline queue / attempt cap**, or (b) **frozen next-item** selection on the last server-known band / pack until sync. Fluency drilling on a frozen pack is acceptable; claiming mastery movement before server commit is not. Full offline adaptive next-item deferred.
2. **Resume flush fixtures.** Pause-hold resume reconcile is quiet mastery credit — celebrationTier fail-closed (none / quiet); no bulk celebration dump. Build exit fixtures must cover: offline provisional without band promotion; resume flush fail-closed; same-key replay returns original ClientView (§22).
3. **4-beat diagnostic contract.** oneFocus (and, where tagged, supporting beats) is **deterministic from Item Bank item / misconception metadata** — not free-form vanity praise. Item Bank owns tags; server feedback assembly sources them; Interface renders; Signal may refine misconception taxonomy later. Generic cheer + celebration flash alone is not a learning claim.

Not reopened: QualifyingEvent spine; soft-state; §18 queue; §21 eval/policy_version; §22 replay + server time; §23 Kids Trust pause/revoke honesty.

Slice 5: Engineering continues. No new hard block beyond existing kid-reachable QE mint gates (§22) and Legal disclosure before offline ships (§23).

25. Supersession: derived fuel counts on the companion payload (2026-09-24)

Source: Samarth dogfood lock (fuel unmissable on home) + Opportunity memo §11d amendment.

Disposition: no-reopen. Supersedes the Slice 5/7 "companion payload has no XP total / home line has no count" wording.

Architecture lock:

1. Child companion/home payload may carry **read-only derived counts**: flame run (from `QualifyingPracticeDay` only), XP total (sum of `QualifyingEvent`-pointed ledger credits), `BuildGoal` pieces (from the one `BuildGoal` projection).
2. These are **projections of the append-only ledger and the QE bus**, computed server-side. No separate counter table the client or another path can increment. No write path from the client.
3. Counts are fuel state, not mastery state. They never feed `Learner State`, progression, or band decisions, and they are not `ClientView` soft-state (no %, no confidence).
4. Parent one-breath stays XP-free (Parent Trust consumes derived narrative only).
5. Opportunity guards hold architecturally: no per-attempt XP ticker, flame counts qualifying days not sessions, XP never decreases, no cross-child comparison.

Not reopened: `QualifyingEvent`-only mint; append-only XP; one `BuildGoal`; soft-state `ClientView`; Parent Trust minimization.

26. Item generation: placement, identity, offline (2026-09-24)

Source: Samarth lock (question variety is a dogfood blocker) + Opportunity memo §11e. **Disposition:** **additive, no-reopen.** Item Bank moves from fixed items to deterministic per-skill templates. Concept Graph node ids, the `QualifyingEvent` bus and soft-state are unchanged.

Architecture locks:

1. **Generator is server-side, inside Item Bank.** Templates, operand draws and answer computation run on the server. Selection stays with Progression Policy, which asks Item Bank for an item at a skill and difficulty step. No LLM in the generation path.
2. **Concrete item identity.** Each issued item has a stable `item_instance_id` and records `template_id`, `template_version`, `difficulty_step`, and the **resolved parameters** (the operands, not just an RNG seed). Replay and the offline eval harness read stored parameters, so they never depend on re-running a random generator.
3. **Attempt log.** Every attempt carries `item_instance_id` + `template_id` + `template_version` + `difficulty_step`, next to `policy_version` and the app build commit SHA (Signal's ask, endorsed). Misconception tag and `oneFocus` / `whyItWorks` resolve from the template version, so feedback stays deterministic.
4. **No-repeat window is server state.** Per child, keep an issued-item record keyed by template + parameter hash. Enforce 7-day exact-repeat exclusion and no back-to-back template within a session at issue time. The client never decides repeats.
5. **Scoring stays server-authoritative.** The server re-computes correctness from stored parameters on submit or sync. Mints, bands and `celebrationTier` come only from that commit.
6. **Offline (fits §24 cap).** Before going offline the client holds a server-issued batch of concrete items, sized to the offline attempt cap. The batch is the frozen pack; the client does not generate items.

Tradeoff: §24 allows a provisional correct / not-yet verdict offline, which means the batch carries enough for the client to check answers. For personal dogfood that is accepted. If Math Sprout is distributed more widely, revisit (options: no offline verdict, or accept the exposure). Unused batch items expire and never count toward the no-repeat window unless attempted.

Not reopened: Concept Graph topology; QualifyingEvent-only mint; soft-state ClientView; §21 eval boundary; §22 replay and server time; §24 cap.

27. Amendment to §22: flame heat states and offline day attribution (2026-09-24)

Source: Opportunity heat definition + Signal ML Locked item 9. **Disposition:** amends §22 server-time rule; no-reopen.

Streak state machine (matches §4.8): heat depends only on how recent the last `QualifyingPracticeDay` is, capped at one qualifying day per calendar day in `child.timezone`. Hot = qualifying day today. Warm = last one yesterday. Ember = one day missed, count holds, practice relights to Hot. Resting (was Dormant) = two or more days missed, run ends, shown as "Flame resting", never 0. XP, attempt count and time never affect heat. The payload sends the state, never a color.

Qualifying-day guards (server-enforced):

1. At least one of the day's K honest attempts must come from Recommended or Challenge. Review alone cannot keep a flame lit.
2. Changing `child.timezone` cannot create a second qualifying day within 24 hours.
3. **Offline day attribution (supersedes strict §22 server time):** an offline attempt counts on its device date only when that date falls between the server time its batch was issued and the server time it synced, evaluated in `child.timezone`. Outside that window, the server's accepted time decides the day. A stale queue cannot backdate an attempt to rescue an ember day.

Still true: server-side derivation only (§25), no client write path, QualifyingEvent-only mint.

§28 Template provenance and evidence eligibility (amends §26)

Status: PR A columns are real. §14 parent authoring is still exploring.

- The PR A template table carries `version`, `provenance (seed | parent | ai_assisted)` and `evidence_eligible` from day one. Parent-authored templates become new rows that pass the same validator. There is no second format.
- Template versions are immutable. Any edit creates a new version. Promotion state is tracked per version, so a revised template starts over in Review.
- `evidence_eligible` is copied onto the `item_instance_id` at issue time. Promotion never turns earlier shadow attempts into Got-it or band evidence after the fact, so replay under `policy_version` stays deterministic.

- Both fields are server-only and are covered by the leak test. The estimator reads the copied value on the server.
- The promotion rule belongs to Signal (ML note, exploring). Validator checks run as code. No model runs at runtime.
- Difficulty prior (exploring): each template version stores `computed_step` (from code-checked features), `parent_grade`, `parent_difficulty` and one `assigned_step` that issuance uses. These live on the immutable template version, not on each attempt, and attempts reach them through `item_instance_id`. If a parent override leaves `assigned_step` different from `computed_step`, the validator step check is recorded as a mismatch rather than rejected, and the shadow comparison settles it. Single-child data may flag a mismatch but never rewrites `assigned_step`; a correction is a new version.

§29 Unreadable answers (amends §22 and §26; applies to PR A whichever UI option Interface picks)

- One answer parser lives on the server and is the authority. The client may pre-check with the same grammar (shared module), but it never decides by itself. The offline provisional path uses the same module, so online and offline cannot disagree about what counts as readable.
- A format reject is not an attempt. It writes no attempt row, emits no `QualifyingEvent`, returns no verdict, no `ClientView` change and no `fuel` earned line, and does not use up the attempt's idempotency key.
- The `item_instance_id` stays open, so the retry is the same issued item. It is already counted as served for the 7-day no-repeat window. The unreadable answer does not add to attempt, qualifying-day or flame counts.
- Format rejects are logged as a separate lightweight event (`item_instance_id`, prompt type, reason), not in the attempt log, so Signal can track the reject rate by prompt type without polluting evidence.
- Format hint (Interface picked retry with hint): the issued item payload carries `answer_kind` (whole | fraction) and a `format_example` that the server chose at issue time and guaranteed is not the answer. The client never compares an example with the answer, because online it must not know the answer. Neither field reveals correctness. `answer_kind` comes from the template's answer type, never from the resolved answer value (a fraction item whose answer comes out whole is still `fraction`), so it never hints at the answer.
- Right value, wrong form (PR #12): this is a readable answer, so it is a scored Not yet attempt, never a `format_rejected`. Routing it through the reject path would give a free retry that confirms the value. The Not yet response carries a typed reason (`wrong_form` plus the required form) as a feedback beat. The client chooses the copy from that reason. The item closes like any Not yet, with no retry on the same `item_instance_id`, and the Lock in beat shows the answer in the required form. It mints nothing and does not count toward a qualifying day.

§30 Session summary (memo §11f, framed; read-only projection)

- The summary is a server-side, read-only projection of the attempt log plus derived fuel state. It is not a mint, does not go through the `QualifyingEvent` bus, and changes no progression. The client

never counts anything itself.

- Session boundary: attempts need a server-issued `session_id` (plus server start time). If attempts do not already carry one, that is the only new column v1 needs. The session ends on an explicit end action or on an idle timeout set in server config.
- Minutes practiced = active time from server attempt timestamps, with each gap capped at a configured idle limit. No client timer.
- Records are "first ever" facts checked against full history under the current `policy_version`: first correct at a higher step on a skill, first Got-it on a skill, first new template type (needs #12), longest correct run of Recommended or Challenge items in a session, and new longest flame. First Got-it and longest flame must be readable from stored band transitions and qualifying days; if band transitions are not stored, the record waits rather than being recomputed on the client.
- The summary is snapshotted once per `session_id` when computed. Refresh or replay returns the same snapshot, so a record never shows twice or changes later.
- Offline: an offline session end shows only the last server numbers (as with fuel). Records are computed after sync and appear once, on the next open, never duplicated.
- Excludes format rejects, legacy items and replays, per Signal's counting rules. The parent view stays XP-free.
- Confirmed (Build, after #10 merged as 6dfe8e0): attempts already carry a required `session_id` pointing at `practice_sessions`, with server `shown_at` and `submitted_at`, so no new session path is needed. The only new storage is the once-per-session summary row keyed on `session_id`, which also records whether the offline "From your last practice" card has been shown, so it appears once.
- First Got-it waits for an append-only band-change log stamped with `policy_version`, which lands in PR B. Longest flame goes ahead only if it's derived from the saved server reward events.

§31 Pool exhaustion (amends §26)

- When the 7-day no-repeat window leaves no fresh item at the child's assigned step, issuance falls back. It never repeats and never shows an empty screen. Switching to another due skill is preferred over stepping up.
- Each `item_instance_id` records `issue_reason` (`normal` | `exhausted_switch` | `exhausted_stepup`). A fallback step-up never changes the child's assigned step. Only the progression rules move steps, so issuance does not become a second decision-maker.
- Signal decides how the estimator, the records and the fast-guess measure treat `exhausted_*` items. The rate of exhaustion events is logged per skill and step as the signal to widen a pool.
- Agreed: `exhausted_stepup` items get `evidence_eligible=false`, copied onto the item at issue time (per §28). They never count toward First Stretch, never move a band and never show the level-up frame. `exhausted_switch` counts as normal evidence and happens silently. `issue_reason` is covered by the leak test. The dogfood deploy gate is a 7-day simulation with zero exact repeats and `exhausted_stepup` under 5% in the heavy case. The pool floor is 10 per template per step and 20 per skill per step.

§31a Pool exhaustion: final rule (supersedes the step-up path in §31)

Settled by Signal in Craft Review; accepted by Build and Opportunity (2026-09-24).

- When a pool runs out, issuance switches skill silently at the child's `assigned_step`. `issue_reason = exhausted_switch`; normal evidence. Only the concept header changes (IA §10).
- If no other skill is available, issue a repeat of the oldest item at the same step. `issue_reason = exhausted_repeat`, logged. The repeat is a **new item_instance_id with evidence_eligible=false copied at issue** (the child has already seen the answer). Only the first attempt on the original instance counts as evidence.
- The step-up path is removed. Nothing is issued from a harder step, so `exhausted_stepup`, its no-Stretch and no-level-up rules, and its 5% threshold no longer apply.
- `issue_reason` values are `normal`, `exhausted_switch` and `exhausted_repeat`. Server-only and covered by the leak test. Fallback never changes `assigned_step` or learner state.
- Floors unchanged: 10 per template per step and 20 per skill per step, counted after engine filters. Step 1 is the CI gate; steps 2 and 3 before PR B.
- Deploy gate: zero repeats in both 7-day simulations (one and two 15-item sessions a day), for every pool including allowlisted ones. A repeat blocks deploy.
- The dogfood readout reports the actual runtime repeat count and switch rate per pool. A real repeat means widen that pool before the test counts (memo §11e).

§31a addendum: switch target and downstream (2026-09-24)

- **Switch target:** on exhaustion, the replacement skill is the next skill in the existing rotation order at its `assigned_step`, skipping exhausted pools. Issuance never picks skills by pool size or any new logic, so the focus selector stays the only skill chooser.
- **Gates (owned by Signal / Interface, recorded here for traceability):** zero repeats plus a switch rate of at most 10% per skill in both rotation runs (one and two sessions a day, 7 days), and at most one switch per 15-item session. A breach means widen the pool or escalate to Samarth as a known dogfood limit. The per-pool stress simulation (b) is kept but is not a gate.
- **Session summary (§30):** skill counts come from items actually issued (using `issue_reason`), not from the planned focus.

§31a addendum 2: template selection precedence (2026-09-25)

- **Exposure is derived, not stored.** "Least recently seen template" is computed from the issued-items log (`item_instance_id` with template id and `shown_at`). No separate per-child template counter, so replay and offline sync cannot drift from what the child actually saw.
- **One selection order, in the single issuance path:** (1) stay on the current skill at `assigned_step`; (2) pick the least-recently-seen template, excluding the template just shown (§11e no back-to-back); (3)

pick an unseen item in that template; (4) otherwise switch to the next skill in rotation order, skipping exhausted pools; (5) logged repeat as last resort.

- **issue_reason values:** `normal`, `template_switch` (step 2 failed: only the just-shown template has unseen items), `exhausted_switch` (skill pool empty at step), `exhausted_repeat`. Both switch kinds count toward the 10% per-skill and one-per-session gates; the readout reports them separately.
- Opportunity's floor stays at 3 distinct templates per skill seen per child per week, checked on a 20-child one-session-a-day run (min / median / max reported).

§31a addendum 3: session extra-slot rotation (#16, 2026-09-25)

- The lane start index fixes only where a session begins, not which skills fill a 15-item session's 4 extra slots. Extra slots may rotate between sessions.
- No mutable per-child cursor. The extra-slot offset is computed at session creation from the child's prior `practice_sessions` (count or last row) and frozen on the new session row beside the lane start. Issuance and replay read it from the row.
- Creating the same `session_id` again must not advance the rotation. Test: two consecutive sessions get different doubled skills, and a retried session creation keeps the same offset.

2B. ML approach note v0 — Math Sprout (ADOPTED)

Source: [ML approach note v0 — Math Sprout \(ADOPTED\)](#)

Status: **ADOPTED v0** (rules-first; 2026-09-21) + LOCKED rulings through 2026-09-25 5:21 PM PT; sections marked EXPLORING are not locked

Owner / author: Signal (AI/ML seat)

Last edited in Notion: Sep 25, 2026, 5:22 PM PT

Compiler's currency note (not part of the Notion source): The pool-exhaustion fallback changed several times on 2026-09-24. The **current rule** is "Corrections (9:00pm PT) → Pool fallback: FINAL, with no step-up" plus the "Last-resort repeat" bullet. It says: switch skill silently, otherwise make a logged repeat with a new `item_instance_id` and `evidence_eligible=false`, and never step up. That rule matches Architecture §31a. The earlier "Running out" sections are superseded. On replayed offline answers, the "Submit-time clamp (5:21pm PT)" rule supersedes the rejection wording in "Replayed offline answers (4:11pm PT)". The 4:19 PM "Update" replaces the `late_replay` allowlist item. The parent-template promotion and difficulty-calibration sections are **EXPLORING**, not locked.

Status: **ADOPTED (v0)** · Product: Math Sprout (Prod 2) · Owner: Signal · Adopted: 2026-09-21 (squad stamps + Samarth web-first) · Sources: [Opportunity memo v1 — Math Sprout \(ADOPTED\)](#) · [Architecture note v0 — Math Sprout \(practice + mastery loop\)](#) · IA/lo-fi: [IA / lo-fi v0 — Math Sprout \(ADOPTED\)](#)

Architecture is ADOPTED. This note is ADOPTED. IA/lo-fi and Build plan are in flight under Facilitator / Build.

0. Verdict (one screen)

v1 is mostly not ML-shaped. Progression, mastery bands, and reward minting should ship as **transparent rules + telemetry** behind stable interfaces. Invite models only when Phase A-B data shows a ceiling (cold start across concepts, noisy prerequisite inference, or integrity features rules can't catch).

Kill for v1 spine: generative AI tutor chat, black-box mastery model as the reward authority, ML that mints XP/pieces, and any parallel “soft score” that bypasses `QualifyingEvent`.

Where intelligence sits (v1): (1) learner-state estimator (rules), (2) next-item policy (rules), (3) attempt-integrity / anti-farm features (rules + heuristics), (4) optional later ML *behind the same interfaces* once baselines exist.

1. Is the problem ML-shaped?

Subproblem	ML-shaped now?	v1 approach	When to reopen ML
Next item / difficulty	No	Rules: stay / remediate / slight level-up from band + recent form	Rules thrash or kids reject “slightly harder” despite clean items
Mastery band (Not yet → Getting it → Got it)	Weakly	Explainable estimator from accuracy, consecutive correct, attempt floor, Review discount	Multi-concept transfer / long-horizon retention needs beyond rules
ConceptProgressTick	No	Early→late session lift on focus concept (accuracy / time-to-correct deltas)	Only if tick is noisy after item-quality pass
HonestAttempt integrity	Mostly no	Deterministic gates: empty, too-fast, identical spam, duplicate sync key	Adversarial farming that beats heuristics
Reward / XP / pieces	Never ML	Server mint from <code>QualifyingEvent</code> only	—
AI tutor chat	Out of spine	Hard kill (Opportunity + Architecture)	Reopen memo + Architecture

Baseline that ML must beat later: the rules estimator + policy above, evaluated on kid preference for fairness *and* measured focus-concept lift — not demo magic.

2. Learner state (Signal owns *how*)

Per child × concept (Architecture lock). Opportunity owns kid-facing band meaning.

v1 estimator (rules)

Inputs (instrument from Phase A):

- Correct / incorrect per attempt
- Latency (ms) + too-fast flag

- Hint use / skip / quit (if present)
- Lane tag: Recommended | Challenge | Review
- Session ordinal within day; early vs late attempts in session

Band sketch (thresholds = placeholders until Samarth + dogfood):

- **Still learning:** recent window accuracy below floor, or $< N$ honest attempts
- **Getting it:** accuracy in mid band with $\geq N$ honest attempts; Review-lane attempts discounted
- **Got it:** accuracy above bar with consecutive-correct streak *and* ≥ 1 Challenge or hard Recommended success (anti-Review-farm)

Kid-facing bands (Opportunity + Interface lock): **Still learning / Getting it / Got it** — no percentages.

UI contract (Interface lock — stamped)

- Client receives **soft state only**: {bandLabel, showConceptChip, celebrationTier} — never raw confidence, never “the AI thinks...”
- Concept chip / Got-it chrome only when `showConceptChip=true` (confidence gate server-side)
- **Review lane:** quiet reduced-XP toast only — **no** LevelUpSlight chrome, no Badge/BuildPiece celebration from Review mints
- 4-beat feedback template is Interface-owned; Signal does not invent alternate copy

oneFocus sourcing (Arch §24 / Opportunity §11c — stamped 2026-09-23)

- **oneFocus** is **deterministic** from Item Bank item + misconception-tag metadata on the scored attempt — not free-form model text, not a second live scorer.
- **Signal owns** misconception-tag **taxonomy refine** (stable tag IDs, mapping wrong→focus cue keys) for the grades 2–4 number sense/ops + fractions neighborhood.
- **Item Bank** stores tags on items; Build serves them; **Interface renders** the 4-beat line from returned keys.
- Soft-state / no-% ClientView unchanged. Not a Slice 5 stop — taxonomy can land with stub pack / Slice 2–3 content.
- Kill: LLM-generated `oneFocus` in production path; parent-visible answer keys via focus copy.

Item variety: taxonomy + difficulty scale (2026-09-24, Opp §11e / Arch §26, Signal-owned)

- **Taxonomy:** skill (11 existing) contains templates (at least 3 genuinely different shapes each), and each template spans difficulty steps. Every issued item has `item_instance_id = template_id + template_version + difficulty_step + operands`.
- **Difficulty scale:** steps 1–3 (entry = 1) for every skill. Each step is defined by features code can check, set per skill: operand size or digit count, regrouping count, where the missing term sits, like vs

unlike denominators, and so on. Steps are hand-assigned and versioned. They are **not** ML- or IRT-calibrated in v1 because there's no data yet. Once dogfood data exists, we check that accuracy falls step by step, and re-tier any step that breaks that order.

- **Legacy items:** the 11 existing fixed items carry forward as v0 static templates at step 1 (Build input, accepted), so history and replay stay valid.
- **Misconception tags:** each template declares bug rules, meaning known wrong computations such as “forgot to regroup” or “added the denominators,” each mapped to a tag. When a wrong answer matches a rule, that tag’s `oneFocus` shows. If it doesn’t match, the template’s default focus shows only when it is real; otherwise the beat is omitted. `whyItWorks` follows the same rule.
- **Progression:** step up after a correct streak at the current step (a config stub), step down after misses at the step. Recommended serves the current step, Challenge serves one step higher, and Review serves the current step or lower.
- **Estimator:** evidence is kept per child, per skill, per step. **Got it** needs success at step 2 or higher from Recommended or Challenge. Step-1-only accuracy caps the band at Getting it. The estimator becomes step-aware, so `policy_version` moves to `rules-v1`, and old attempts replay under `rules-v0`.
- **Instruments:** false Got-it, mint rate vs attempt volume, and Review share of XP are sliced by `difficulty_step` and build SHA. Add an exact-repeat rate (target 0 within 7 days) and a step-up count per skill practiced, which are Opp §11e’s success signals.
- **Kill:** LLM-generated items; ML difficulty calibration before data exists; a client-side generator.
- **Moving from `rules-v0` to `rules-v1` (Build plan §11.9, gates PR B):** all v0 evidence counts as step-1 evidence. A policy change never visibly lowers a band. A v0 Got it keeps showing as Got it but is marked `provisional_v0` internally, and that child starts the skill at step 2 in Recommended. It becomes confirmed with the first step-2+ success from Recommended or Challenge. It can drop only through normal v1 evidence, meaning misses at step 2, so any drop comes from the kid’s performance and never from the migration. Re-estimating at migration mints nothing: no `MasteryBandTransition`, no fuel. Every other band carries over as-is, starting at step 1. The v0 replay stays under v0.
- **Fraction answers:** equivalent forms are accepted ($\frac{2}{4}$ for $\frac{1}{2}$, improper vs mixed) except on templates whose skill is simplifying or converting, which require the target form. The bug-rule collision check (Craft must-fix 6) compares values after canonicalizing, so a bug output equivalent to the right answer is rejected at draw time.

Contracts for Build

- `LearnerState.estimate(conceptId)` → `{band, confidence, recentForm}` (server-internal)
- `LearnerState.toClientView(...)` → `{bandLabel, showConceptChip, celebrationTier}` (web client)
- Attempt feedback keys: `whatWentWell` · `oneFocus` · `tryNext` resolved from item/misconception metadata (+ Interface copy map)

- Persist `policy_version` on attempt/session (stub `rules-v0` OK in Phase A; required before Phase B threshold claims)
- Pure function of stored attempts + policy version — replayable, A/B-able
- Never write XP from this path; only emit events that Reward may consume

3. QualifyingEvent thresholds (Signal proposal — stamp with Opportunity)

Architecture's taxonomy stays. Numeric stubs for Phase A-C (replace after dogfood; do not invent a second taxonomy):

1. **HonestAttempt** — submitted; not empty; latency $\geq L_{\min}$; not duplicate `attempt_idempotency_key`; not identical-answer spam window.
2. **ConceptProgressTick** — within a completed session on focus concept: late-third accuracy – early-third accuracy $\geq \delta$ or first→best continuous score lift $\geq \delta$ (Signal will lock δ after N sessions of data).
3. **MasteryBandTransition** — estimator crosses band boundary with confidence $\geq c_{\min}$ (else “pending”).
4. **LevelUpSlight** — policy-approved only; never from Review-lane alone.
5. **QualifyingPracticeDay** — ≥ 1 session with $\geq K$ HonestAttempts (K small, e.g. 5) on calendar day in child TZ.
6. **Badge / BuildPiece** — mapped only to band transitions, LevelUpSlight, or qualifying streak milestones — **never** raw count.

Anti-events (no full mint): rapid guessing, Review overuse, below-floor sessions, sync retries.

@Build: these are executable rules; @Opportunity: stamp kid-facing meaning of bands; @Interface: show uncertainty + ember recovery without implying model judgment.

4. Eval bars & instrumentation (Phase A must ship)

Without these, we cannot later claim ML beat rules.

- Attempt log: concept, item, difficulty, lane, correct, latency, integrity flags, session id, idempotency key
- Session: focus concept, start→first-attempt, completion, quit point
- Choice mix: Recommended / Challenge / Review rates
- Band transitions + false Got-it rate (Review-heavy paths)
- Parent narrative open (trust) — not an ML metric, but gates keepable product

Demo vs production: any future model demo must be labeled DEMO and must beat the rules baseline on held-out kids/sessions for fairness preference *and* focus-lane lift.

5. Challenges / stamps

Stamp (Architecture)

- QualifyingEvent as joint contract — **yes**
- Rules-first progression behind swappable estimator interface — **yes**
- No ML required for Phase A — **yes**
- Reward never from raw volume / time-on-app — **yes**

Locked (squad stamps 2026-09-21; neighborhood amended same day)

1. **Review dual-cap** — sessions/week ceiling + no LevelUpSlight / full Badge / BuildPiece from Review; reduced XP only; quiet toast, no LevelUp chrome.
2. **Got-it / LevelUpSlight** require Recommended or Challenge evidence.
3. **Soft-state UI** — no raw confidence to client; no “AI thinks” framing.
4. **Web-first client** — browser-hosted practice UI; no on-device model for v1; latency budget = typical web RTT to hosted backend for score + next-item (no streaming inference UX).
5. QualifyingPracticeDay K remains a **config stub** until dogfood.
6. **v1 concept neighborhood (Samarth amendment)**: grades 2–4 number sense / operations **and fractions** — estimator + Phase A instrumentation node IDs include both. Prior “fractions out of spine” **superseded**. Still no ML-in-spine.
7. **oneFocus / misconception tags (2026-09-23)**: deterministic from Item Bank metadata; Signal owns taxonomy refine; Interface renders (Interface stamped render side); soft-state unchanged. Not a Slice 5 stop.
8. **Kid-visible fuel counts (2026-09-24, F1 v2; Opp §11d / Arch §25)**: flame / XP / pieces shown to kid are read-only derivations from QE ledger + QualifyingPracticeDay + BuildGoal projection. Counts are fuel state only — never an estimator input or band signal. Instrument watch after ship: QE mint rate vs attempt volume and Review share of XP, before vs after the counts became visible (farming pressure check).
9. **Flame heat states (2026-09-24, Opp definition)**: hot/warm/ember/resting come only from how recently the last QualifyingPracticeDay was. Farm guards: a QPD needs at least 1 of its K HonestAttempts from Recommended or Challenge, so Review alone can't light the flame; a `child.timezone` change can't produce a second QPD within 24h; offline attempts count for a day only between batch issue time and sync, with no backdating before the batch was issued. Log heat transitions and each QPD's lane mix.

Non-blocking

Avatar/warrior BuildGoal skin — product lock; ML-agnostic. Parent-driven LLM graph growth stays parked v2+ (Opportunity §13).

6. Implications by role

- **Opportunity:** Keep rules-first bias. Stamp §3 event stubs or counter with alternate numbers. Own band labels.
- **Coherence:** Architecture v0 shape is ML-coherent; keep estimator behind Learner State boundary.
- **Build:** Implement rules estimator + integrity gates + idempotency; leave `MasteryEstimator` interface for later swap. No Build plan until IA also adopted.
- **Interface:** 4-beat + soft-state view + Review quiet toast — stamped; `oneFocus` render side stamped (keys from Item Bank / Signal taxonomy).
- **Craft:** Later — ledger + integrity as security-sensitive; eval fixtures for farm patterns.

7. Decision log (v0 · ADOPTED)

Decision	Choice	Rejected
v1 intelligence	Rules + telemetry + swappable interface	ML-first mastery / next-item
Rewards	QualifyingEvent only; never model-minted	Soft score / engagement model XP
Tutor chat	Out of spine	Chat as pedagogy
Review	Dual-cap + quiet reduced-XP toast; no LevelUp chrome	Unlimited easy XP / Review LevelUp
Client	Web browser → hosted backend; no on-device inference	Native / on-device model path
UI state	Soft labels + celebrationTier only	Raw confidence % to kid
When ML enters	After rules baseline + ceiling evidence	Ship model to “absorb” fuzzy product
<code>oneFocus</code> source	Item Bank misconception tags → deterministic keys; Signal taxonomy	LLM free-text focus in prod

Next: Signal refines misconception taxonomy with Slice 2 stub pack; numeric stubs post-dogfood. Eval harness = offline attempt-log consumer (Build owns wiring; Signal owns baseline definition).

EXPLORING (not locked): promotion rule for parent-added templates (Opp §14, Arch boundary) — 2026-09-24

Status: exploring. Runs after dogfood hold and PR A. No build yet.

- **Starts ineligible.** New parent-added or AI-assisted template: `evidence_eligible = false`. Issued only in the Review lane, Review-level XP inside the existing Review dual-cap, never Got-it or band

evidence. It can't create a `QualifyingPracticeDay` by itself, because a QPD needs at least one Recommended or Challenge attempt.

- **Authoring gate (code, before the parent sees it):** the step features computed from generated samples must match the declared `difficulty_step`, and the operations used must be in the declared skill's allowed set. Otherwise the template is rejected. This sits alongside Coherence's validator: every sample computes an answer, and bug rules produce wrong answers that differ from the right one.
- **Shadow gate (behavior, after live):** compare the same child's accuracy on the candidate with their accuracy on seed templates at the same skill and step. Promote when all of these hold: at least 20 attempts across at least 3 days, accuracy within about 15 points of seed, and at least a third of misses matching a declared bug rule. Much higher accuracy means the step is set too low. Much lower means the step is too high or the skill is wrong. Few bug-rule matches mean oneFocus would be filler. Any failure keeps it in Review, and the parent can keep it as practice.
- **Demotion:** the same checks keep running after promotion. Drifting outside them sends the template back to Review.
- **Honesty:** the thresholds are config stubs, not calibrated values. With one or two kids the data is thin, so the code gate and parent approval do most of the work, and the shadow gate is a sanity check.
- **Instruments:** every existing instrument is split by provenance (seed / parent / ai_assisted), so candidates never mix into seed numbers.
- **Build (PR A):** `version`, `provenance`, and `evidence_eligible` columns from day one, which I endorse. The estimator reads `evidence_eligible` on the server only. `provenance` and `evidence_eligible` go in the ClientView leak fixture.

EXPLORING amendment: difficulty calibration for parent-added templates (Samarth input; Opp §14; Arch §28) — 2026-09-24

- **Supersedes the authoring-gate line above for steps only.** A step mismatch is **flagged, not rejected**. The template version stores `computed_step` (from code-checkable features), the parent prior (worksheet grade plus per-question Easy/Medium/Hard), and one `assigned_step` that issuance reads. A skill mismatch (operations outside the skill) and features outside every step's range still reject.
- **Now, prior plus features:** when the parent prior and `computed_step` agree, assign that step. When they disagree, flag it on the approval screen. The parent's override sets `assigned_step`, and the code step wins if the parent skips.
- **Mismatched templates in shadow:** the shadow gate compares the child's accuracy with seed templates at **both** `assigned_step` and `computed_step`. It promotes only when accuracy fits `assigned_step`. If accuracy fits `computed_step` instead, it stays in Review and the approval screen suggests a new version at that step. It never auto-rebases.
- **One-kid confound guard:** with one child, correct rates mix ability with item difficulty. Single-kid data may **flag** a suspected mis-rated template but never rewrites a step. A correction is always a

new version (Arch §28).

- **Later, statistical calibration:** needs many learners or an ability-adjusted model (IRT-style). The no-IRT-until-data lock is unchanged. Instruments: every existing slice also splits by prior agreement (agree / mismatch / no prior), read through `item_instance_id` to the template version.

Answer outcomes: wrong form, unreadable answers, blanks (LOCKED 2026-09-24, PR #12)

Right value, wrong form (`form_mismatch`). This is its own evidence tag on the attempt row. On `require_form` templates the form is the skill being tested, so on `rules-v0` it scores as a full miss for that skill, with no partial credit and no change to the bands. It mints nothing and stays blue. The item closes with no retry, because once the Lock-in line shows the answer, a second try only shows the child can copy it. The child sees `reason: { kind: "wrong_form", required }` and Interface's copy. `form_mismatch` itself stays internal and is covered by the leak test. PR B (`rules-v1`): repeated `form_mismatch` on a skill routes the child to form practice (for example, simplifying to lowest terms), not the general re-teach a wrong value gets.

Mixed-number templates. The generator skips operand sets whose answer comes out whole, so a correct `1` is never marked wrong. Improper-fraction templates still reject `1`.

`answer_kind` comes from the template's answer type, never from the value of a particular answer. A fraction item whose answer comes out whole still gets `fraction` and the `"/` keyboard, so the keyboard doesn't reveal the answer.

Unreadable answers (Arch §29). An unreadable answer is not an attempt: no attempt row, no evidence, not a miss, nothing minted. It doesn't use up the retry key, and the first answer that parses scores as the child's first attempt. Rejects go to `answer_format_rejects` with `item_instance_id`, `template_version`, `provenance`, `answer_kind`, `build_sha`, `policy_version` and `reject_seq`, and never the raw input. This table is an input-quality check only and never feeds the estimator, the bands or fuel. The rate is rejects divided by served items, per template and per prompt type. Joining on `item_instance_id` shows whether the next answer parsed. This supersedes my earlier ask to store `unparseable` on the attempt row.

Blanks. Unchanged on `rules-v0` (`empty_answer`: no XP, not a miss for remediation, but it still takes a slot in the mastery window). Any change comes in PR B under `rules-v1`, with a `policy_version` bump.

Lowest-terms rulings and the reject-rate headline (LOCKED 2026-09-24, Craft questions on #12)

- **Step 1 on lowest-terms templates draws proper fractions only.** An answer above 1 adds a second idea, so it belongs at step 2 or higher.
- **Lowest_terms means the fraction part can't be reduced any further.** Both improper and mixed answers are accepted: for an answer of $3/2$, both $3/2$ and $1\ 1/2$ are correct, and $1\ 2/4$ is `form_mismatch`. Templates that need a specific form use `improper` or `mixed`. False misses do more damage to the evidence than a lenient form check does.
- **Whole-number values:** `1` and $3/1$ are both correct on lowest terms. The Lock-in line shows the stored canonical form.

- **formatRejectRates headline:** distinct rejected items divided by items served, per template id, version, provenance and answer kind, with total rejects as a second number. A large gap between the two means one child hit the same item repeatedly, not that the template's input format is bad.
- **Tests:** step 1 never draws an improper fraction, $1 \frac{1}{2}$ is correct on lowest terms, $1 \frac{2}{4}$ is `form_mismatch`, $\frac{3}{1}$ is correct.

Session summary stat rules (LOCKED 2026-09-24, Opportunity §11f, Architecture §30)

What counts: scored attempt rows only (unreadable answers never create a row). Only the first scored attempt on each item counts, so retries and replays are dropped. Legacy v0 items don't count.

- **Minutes practiced:** the sum of `submitted_at` minus `shown_at` for each item, capped at 2 minutes per item, rounded down, and hidden if under 1. A plain fact, never a record.
- **Questions practiced:** the number of distinct items with a counted attempt. A plain fact, never a record.
- **New question type:** the child's first attempt ever on a `template_id`. A new version of a template they've seen doesn't count. Needs #12.
- **First Stretch:** a correct first attempt at a step higher than the child's best correct step so far on that skill, using the step saved on the item when it was issued.
- **Longest correct run:** correct first attempts in a row on Recommended and Challenge items. A wrong answer or `form_mismatch` ends the run; an unreadable answer neither adds to it nor ends it. Counts as a record only if the child has at least one earlier session and the run is at least 3.
- **First Got-it:** DEFERRED. Bands are recomputed today, and the rules-v1 change in PR B would rewrite past bands. Needs an append-only band-change log stamped with the scoring-rules version, added in PR B.
- **Longest flame:** allowed only if derived from saved server reward events, not re-derived from the flame rules. Build to confirm what the flame reads.
- **Fast guess (success signal):** an answer submitted less than 2 seconds after the question was shown. The cutoff is uncalibrated and will be recalibrated from dogfood timing data. It doesn't gate any record in v1.
- **Longest flame (update, 2026-09-24):** APPROVED, pinned on the saved summary row. The best run so far and the new run are stamped with `policy_version` and never recomputed. A record only counts against a best run saved under the same `policy_version`. When the flame rules change, the best run is recomputed once under the new rules and saved as the new starting point, and no record is shown that session. Test: a `policy_version` change shows no record and saves the new starting point.

Step pool minimums (LOCKED 2026-09-24, Craft delta pass on #12)

The step-1 pools at `d0970cd` (mixed 1, improper 2, scale 2, lowest 7) break the 7-day no-repeat rule within days. Samarth ruled repeats a dogfood blocker. This blocks the dogfood deploy, not merging #12.

- **Minimums:** at least 10 distinct instances per template per step, and at least 20 per skill per step across its templates. Widen the operand ranges that fit the step; never draw from a harder step to fill a pool.
- **CI check:** count each template's step pool after the engine filters (lowest-terms proper-only, mixed whole-answer skip) and fail under the minimum.
- **Running out:** repeat the instance seen longest ago at the same step and log it as a repeat. Never quietly move the child to a harder step, because harder items then show up as misses in their record.
- **Basis:** about 15 items a session with an 11-slot skill rotation works out to roughly 10 to 14 items per skill per week for a daily user. The minimums are an estimate, not measured, and get rechecked against the dogfood repeat log after week 1.

Running out of a pool: fallback order and evidence (LOCKED 2026-09-24; SUPERSEDES the 'Running out' bullet in Step pool minimums)

The earlier rule (repeat the instance seen longest ago at the same step) is superseded, because Samarth ruled repeats a dogfood blocker. The new order follows Architecture §31: switch to another skill at its assigned step first; step up only when no other skill is available; never repeat an item; never change `assigned_step`.

- `issue_reason=exhausted_switch`: normal evidence.
- `issue_reason=exhausted_stepup`: `evidence_eligible=false`, frozen at issue. Never counts toward First Stretch and never moves a band. Rewards for a correct answer are the same as any item. Test: a correct `exhausted_stepup` item leaves the band unchanged and shows no Stretch record.
- **Floor:** at least 10 per template per step (`P00L_FLOOR`) and at least 20 per skill per step across templates, counted after the engine filters. Step 1 is the dogfood gate; steps 2 and 3 get the same floor before PR B.
- **Deploy simulation:** one session of 15 items a day for 7 days, plus a heavy case with two sessions a day. Pass means zero exact repeats and `exhausted_stepup` under 5% of items in the heavy case. If step-ups exceed that, widen the pool.

Running out of a pool: final rule (LOCKED 2026-09-24; SUPERSEDES the step-up fallback above)

The order is: switch to another skill at its assigned step; if no skill is eligible, repeat the item seen longest ago at the same step and log it. Issuance never pulls from a harder step and never touches `assigned_step`. The `exhausted_stepup` value is removed.

- `issue_reason=exhausted_switch`: normal evidence.
- `issue_reason=exhausted_repeat`: counts as evidence only the first time the child sees that item, which is the same rule that drops retries and replays.
- **Deploy gate:** a 7-day simulation at dogfood volume (one session of 15 items a day, plus a heavy case with two sessions a day) with zero exact repeats and zero `exhausted_repeat`.

- **Allowlist:** being on it excuses a pool from the size check only, not from the simulation. An allowlisted pool that can't pass the heavy case blocks the deploy until it's widened.
- **Pool counting:** the minimums stay (at least 10 per template and at least 20 per skill at every step). Count the pool the engine actually draws from, per template version and step, snapshotted next to the content hash.

Running out of a pool: FINAL (LOCKED 2026-09-24 9pm PT; SUPERSEDES both earlier fallback sections)

Agreed in Product Squad (Build pool PR, Interface IA §10 additions). This applies to both rooms.

1. Switch to another skill at its assigned step (`exhausted_switch`): normal evidence, and the switch is silent.
 2. If no other skill is available, give an item from the next step up (`exhausted_stepup`): `evidence_eligible=false`, frozen at issue. No band movement, no Stretch record, no level-up frame. The difficulty badge stays accurate and rewards are unchanged.
 3. If no higher step exists, repeat the item seen longest ago (`exhausted_repeat`) and log it. It counts as evidence only the first time the child sees that item.
- **Gate:** 7-day simulations at one and two sessions a day, with zero repeats and `exhausted_stepup` under 5% in the heavy case. Floors: at least 10 per template and at least 20 per skill per step, counted after the filters. Step 1 is the CI gate; steps 2 and 3 get the same floor before PR B. The allowlist excuses a pool from the size check only, not from the simulation.

Corrections (2026-09-24 9:00pm PT)

Wrong-form rewards (clarifies Answer outcomes). Where the note said `form_mismatch` "mints nothing", it meant that none of the rewards for a correct answer are minted. On rules-v0, a `form_mismatch` scores exactly like a wrong answer, including effort XP and credit toward the qualifying day and streak (Craft's #13 must-fix 1). Any different reward needs a written decision and a rules-v1 economy PR.

Pool fallback: FINAL, with no step-up (SUPERSEDES the 9pm 'FINAL' section above). 1) Switch to another skill at its assigned step (`exhausted_switch`), silently, as normal evidence. 2) If no skill is eligible, repeat the item seen longest ago at the same step (`exhausted_repeat`), logged, and counted as evidence only the first time the child sees that item. Issuance never draws from a harder step, so `exhausted_stepup` and its level-up-frame rule are removed. Gate: 7-day simulations at one and two sessions a day with zero repeats; the allowlist excuses a pool from the size check only. The rooms flipped on this three times; any further reopening goes to Samarth as an escalation.

- **Last-resort repeat (aligned with Architecture §31a, 2026-09-24):** a repeat gets a new `item_instance_id` with `evidence_eligible=false` set when it's issued, so an attempt on a repeat never counts as evidence. Only the first attempt on the original item counts. This tightens the earlier "evidence only the first time the child sees it" wording. The dogfood readout reports real repeats and switch rates per pool (Opportunity §11e).

#14 review: switch-rate gate (2026-09-24 10:30pm PT)

- **Simulation (b)** gives 210 items to each skill pool on its own. It passes through `exhausted_switch` whenever any other skill has fresh items, so it can't fail on repeats. Keep it as a stress test only.
- **Gate added:** a heavy rotation run (two sessions of 15 items a day for 7 days on the 11-slot rotation) alongside the one-session run. Both need zero repeats and a switch rate of at most 10% per skill. A pool over that gets widened, or goes to Samarth as a known dogfood limit. The likeliest to cross it is comparing unit fractions at step 1 (19 items: 3 templates of 6 plus the v0 seed).
- **Known evidence skew (not blocking):** at step 1, finding an equivalent fraction has improper 45, scale 10, lowest 7 and mixed 2. Once the small templates run out, step-1 evidence mostly measures the improper form. Check the template mix in the dogfood readout.

#14 rulings (2026-09-24 10:35pm PT)

- **A repeat mints nothing that comes from evidence.** An item with `issue_reason = exhausted_repeat` creates no band movement, `MasteryBandTransition`, `BadgeMilestone`, `ConceptProgressTick` or `Stretch` record. XP and fuel follow the normal rules-v0 path, the same as `form_mismatch`.
- **Session summary:** stats and records skip `exhausted_repeat` items. A repeat's new item id must not count as a fresh first attempt.
- **Step-1 improper fractions:** the whole part of the answer is capped at 3 (denominators unchanged), shipped as a new template version with counts re-snapshotted. The larger answers (whole part 4 to 6) belong to step 2. If the pool falls under the floor of 10, report the number rather than loosening the cap.

Template weighting (2026-09-25 12:12am PT)

- **Equal weight per template.** Within a skill and step, issuance picks the template this child saw least recently among templates that still have unseen items, and breaks ties with the child's seed. Pool size reflects how many operand combinations a template has, not how important its format is, so no weighting by pool size.
- **Dogfood readout:** report each template's share of draws per skill. Any template over about 60% of a skill's draws in real use comes back to Signal.

What counts as seen (2026-09-25 12:15am PT)

- An item counts as seen only once it has been shown to the child (`shown_at` is set). Being handed out is not enough.
- An abandoned prefetch that was never shown does not count toward the 7-day no-repeat window, the least-recently-seen template choice, the switch rate or the 3-template floor, and it is never evidence.
- This is a condition on bringing back any client prefetch or offline batching (PR B / #15). No change to #14 or dogfood, since main has no prefetch.

#16 gate result (2026-09-25 2:12am PT)

- **ML gate met at d812e4b**, as long as Craft's code check confirms the switch counts are real. All 8 lane runs (recommended, challenge, index 6 and review, each at 1 and 2 sessions a day, through `startPracticeSession`) had zero repeats, 0% switches, at most 1 switch per session, and at least 3 templates per skill.
- The extra slots rotate with offset $(\text{overflowSlots} \times \text{earlier session count}) \bmod \text{skillCount}$, a step of 4, so back-to-back sessions double different skills and the worst skill in a 14-session week gets 19 requests.
- **Thinnest pool**: comparing unit fractions at step 1 gets 19 requests against a 19-item pool at 2 sessions a day. A 20th request is an `exhausted_switch`: silent, still counts as evidence, no repeat. That's 1 switch in 20 (5%). It would take about 3 extra sessions a week to go over 10%. The dogfood readout tracks requests per child-week against the pool, and this is the first check on week-1 data.
- Equivalent fractions: at most 19 requests against a pool of 41, and it never gets down to one fresh template. The §11e back-to-back question doesn't come up at these volumes.

Replayed offline answers (2026-09-25 4:11pm PT, for #15)

- **Response time comes from the device**. Response time (used by the 2s fast-guess cutoff and the 2-minute-per-item cap) is measured on the device when the child answers. It is stored on the queued entry and sent unchanged on every retry. It is never calculated from when the server receives it.
- **Late replays don't change the band out of order**. A parked answer can reach the server after newer attempts on the same skill. When it does, it is still saved and still earns XP, fuel and streak credit under the normal rules, but it is marked `late_replay` and does not count toward band changes. If nothing newer exists for that skill, it counts normally.
- An answer parked on a timeout stays parked, the same as after a network failure (Craft's must-fix). The idempotency key makes sure a replay the server already saved is credited only once.
- Tests: the saved response time equals the device-measured value after a timeout followed by a successful replay; a late replay does not move the band; an on-time replay does.

Late replays and progress signals (2026-09-25 4:19pm PT, for #15)

- **Concept-progress tick**: `late_replay` behaves exactly like the other non-evidence attempts (for example `exhausted_repeat`), using the same evidence check. It gets no separate switch. The principle: anything the child reads as learning progress follows evidence, and anything that counts effort follows rewards. Test that `late_replay` behaves the same as `exhausted_repeat` for the tick.
- **What must be in before kids use it (Craft's sort)**: `late_replay` saved with evidence 0 (a reason column added by migration, band update skipped), response time measured on the device with server bounds, a timeout keeps the answer parked, and only one queue flush runs at a time. The `late_replay` XP/streak allowlist goes to PR B if it takes more than one line.
- **PR B**: order band evidence by the server's `issued_at` instead of the device's `submitted_at`, so the band never depends on the device clock.
- **Update (4:19pm PT), replaces the allowlist item above**: Craft checked `lib/qualifying-bus.ts` at `cda8b32`. The progress tick already requires `countsForBand !== false`, and `HonestAttempt` XP and

`QualifyingPracticeDay` are minted regardless of it. So `late_replay` just passes `countsForBand: false` (as `exhausted_repeat` does). It still earns XP and streak credit, the tick is suppressed, and the band doesn't move. No allowlist and no special switch are needed, so nothing about `late_replay` goes to PR B.

Response-time cap (2026-09-25, 4:44pm PT)

LOCKED for #15 must-fix 3. The server clamps stored response time at 120 seconds. This is the same limit as the session-summary minutes rule (2 minutes per item), and both use one shared named constant so they can't drift apart. The value is clamped, not rejected. An answer that took longer than 2 minutes is still a normal attempt and still counts as band evidence. The `submittedAt` bounds are separate and stay as rejections: a time in the future beyond the skew allowance, or a time earlier than `issued_at`. Craft asked for boundary tests at 119, 120 and 121 seconds and at a very large value. This clarifies the earlier '#15 replayed answers' ruling, which said the time is capped but didn't give a number.

Submit-time clamp (2026-09-25, 5:21pm PT)

LOCKED for #15 must-fix 2B. This SUPERSEDES the rejection wording in 'Replayed offline answers (4:11pm PT)'. The server never rejects an answer for clock reasons.

- The stored `submitted_at` is the device's `submittedAt` held inside the range from server `issued_at` to server receive time: the later of the device time and `issued_at`, then the earlier of that and the receive time. `submitted_at_clamped` is set whenever either limit applies.
- Only a genuinely malformed value is refused, and it gets a typed error code rather than a retry.
- The streak and XP day is worked out from the stored, clamped time in the child's timezone. Tests use clocks 20 seconds slow, 10 minutes slow and 3 minutes fast, and the late-replay day-credit test uses the stored time.
- Band evidence is not affected. Lateness and evidence order use server `issued_at`. Response time is the device's own shown-to-Check difference, clamped to between 0 and 120 seconds, so a wrong clock setting cancels out. A clamped attempt is still full evidence.
- The flag only feeds the dogfood readout. If more than about 5% of one child's attempts are clamped, we look at that device. It gates nothing.

3. Design Notes

3A. IA / lo-fi v0 — Math Sprout (ADOPTED)

Source: [IA / lo-fi v0 — Math Sprout \(ADOPTED\)](#)

Status: **ADOPTED v0** (2026-09-21) + additive supersessions §9–§13 (2026-09-24)

Owner / author: Interface (Design seat)

Last edited in Notion: Sep 24, 2026, 9:02 PM PT

Compiler's currency note (not part of the Notion source): The page already carries dated "Superseded Sep 24, 2026" callouts in the places they apply. §9 (F1 v2 child home), §10 (session summary card), §11 (pool exhaustion), §12 (answer-input rules) and §13 (option B fuel strip + color map) are current. They supersede v0 frame F1, parts of F2, the J4 glance-only line, the §3 surface tree and parts of the §4 feedback frame.

Status: **ADOPTED (v0)** · Product: Math Sprout (Prod 2) · Owner: Interface · Adopted: 2026-09-21 · Sources: [Opportunity memo v1 — Math Sprout \(ADOPTED\)](#) · [Architecture note v0 — Math Sprout \(practice + mastery loop\)](#) · [ML approach note v0 — Math Sprout \(ADOPTED\)](#)

Architecture + ML + IA/lo-fi are now all adopted. @Build's default next move is Build plan v1 (Craft Review can engage after).

Distribution lock (Samarth): web app first — responsive web journeys; no native-app chrome in v1.

0. Thesis (one screen)

Child cognition wins when every screen answers three questions fast: **What am I practicing? Am I winning at it? What happens next?**

Parent cognition wins when one glance answers: **Did practice happen? On what? Did skill move — honestly?**

Gamification appears only as *earned consequence* of QualifyingEvents — never as a parallel circus that steals attention from the attempt loop.

v1 IA spine: Session (attempt loop) is home. Companion/build, streak ember, and XP are *side effects made visible*, not destinations that replace practice.

1. Stamps (Architecture + Signal → UX)

Call	Interface stamp	UX implication
QualifyingEvent as joint contract	Yes	Progress chips, XP pops, badge unlocks, build pieces, ember heat — same event family. No soft "engagement" celebrations.

Call	Interface stamp	UX implication
Difficulty visible + boundary-only choice	Yes	Difficulty badge always on item. Choice UI only at session/set boundary — never per question.
Review dual cap (sessions/week + no LevelUp/Badge)	Yes	Review labeled as confidence rebuild; remaining uses shown. Reduced XP toast — never full level-up chrome.
Got-it requires Recommended/Challenge evidence	Yes	Never celebrate Got-it / LevelUp after Review-only path. Soft “still learning” when confidence low.
Ember streak (Hot→Warm→Ember→Dormant)	Yes	Warm recovery copy + rekindle celebration. Zero streak-shame. Parent never sees “broken.”
No “the AI thinks...”	Yes	Kid-facing = practice outcomes + concept names. Rules/ML stay behind the glass.
One BuildGoal	Yes — avatar/warrior skin (Samarth)	One active BuildGoal; pieces from QualifyingEvents only. Presentation = avatar/warrior power-up. Still a side effect of practice , never a second home.
Web-first (Samarth)	Yes	Responsive web layouts; touch-friendly hit targets; no native nav/tab-bar chrome. Offline queue UX can exist later without native shell.
Soft-state UI contract (Signal)	Yes	Client receives <code>bandLabel</code> • <code>showConceptChip</code> • <code>celebrationTier</code> only — never raw confidence or “AI thinks...” copy.
Review chrome (Signal)	Yes	Quiet reduced-XP toast only — no LevelUp / Badge / BuildPiece chrome from Review.
Pause = hold (Opportunity + Arch §24)	Yes — shippable Slice 2	Parent-visible waiting state + quiet child resume (no celebration dump). Drop-on-pause rejected as default; fallback only if hold truly unshippable.
<code>oneFocus</code> diagnostic (Opportunity §11c + LS)	Yes	Beat 2 of 4-beat feedback is diagnostic from item/misconception tags — not vanity encouragement. Client renders server-assembled string; no local rewrite to praise.

2. Journeys (IA)

J1 — Child: focused practice session (Phase A home)

1. Open → **Today's focus** (concept name + plain kid line, not grade)
2. Optional **lane choice** only if boundary moment (else straight to Recommended)
3. Item with **visible difficulty** + attempt
4. Feedback frame (locked template below) → next item or session complete
5. Session complete → only *earned* reward moments (if QualifyingEvents fired) → home

Superseded Sep 24, 2026 by §10. The session summary card now replaces the simple end-of-session moment and appears after the last item and before Back home.

Task success: child can name the focus concept after the session; never lost in menus.

J2 — Level-up / slightly harder (Phase B)

Boundary celebration: **what just got a little harder** (same concept harder params *or* adjacent unlock) + one Challenge invite. Not a random harder quiz from elsewhere.

J3 — Ember recovery (Phase C)

Home shows ember (not a red X). Tap → short warm path into a QualifyingPracticeDay session. Rekindle celebration on return day only after qualifying attempts — not on open alone.

J4 — Companion / build (Phase C, secondary)

Home glance: **avatar/warrior** power progress + last piece earned. Deep view: pieces map to mastery milestones. Companion reacts to QualifyingEvents; **no nag** on miss days. Practice CTA stays primary — warrior progress is glance-only on home.

Superseded Sep 24, 2026 by §13. Fuel, including warrior pieces, now also shows on the practice screen as a slim progress strip, so it is no longer glance-only on home. The practice CTA stays primary.

J5 — Parent trust (Phase D, parallel)

One-breath card: minutes practiced · focus concept · band movement (Not yet / Getting it / Got it). Consent / pause / revoke reachable in ≤ 2 taps. **Pause hold:** parent sees calm waiting state ("Practice paused / waiting to sync"); child quiet-resumes into same focus — no celebration dump, no shame. **No raw wrong-answer dump.**

3. Information architecture (surfaces)

- Child Home
 - └ Focus CTA (primary)
 - └ Ember / streak (glance, warm)
 - └ Companion goal (glance)
 - └ Concept badges (secondary)
- Session
 - └ Focus header + difficulty badge
 - └ Item → Attempt → Feedback
 - └ Progress within session (quiet)
 - └ Boundary: lane choice (rare)
- Reward moments (modals/toasts – interrupt sparingly)
 - └ XP / badge / piece (QualifyingEvent only)
 - └ LevelUpSlight explanation
- Parent Home
 - └ One-breath progress
 - └ Consent & controls
 - └ (no child session chrome)

Superseded Sep 24, 2026 by §13 in part. Under option B the Session surface also carries the slim fuel strip (flame, XP, pieces) and a reward line on each correct answer that earns a reward, so fuel no longer lives only on Child Home.

Cut from v1 nav: shop, leaderboards, multi-goal picker, tutor chat, grade picker as primary.

4. Feedback presentation (Interface lock)

Kid feedback after an attempt is always four beats — Opportunity §4 “encouragement as design”:

1. **What went well** (specific, short)
2. **One focus** (single **diagnostic** correction, server-assembled from item/misconception metadata — never a lecture, never vanity praise). Aligns Opportunity §11c + Arch §24.
3. **Try next** (next item affordance)
4. **Concept chip** — only if band/confidence allows; else soft “still learning *concept*”

Partially superseded Sep 24, 2026 by §12. For right-value, wrong-form fraction answers, Try next is intentionally hidden. Unreadable answers get a single hint line and no verdict, so they do not use this four-beat frame at all.

Never: shame, peer compare, raw model scores, “AI says you’re...”

Soft-state contract (Signal — stamped): client consumes `bandLabel` $\in$ {Still learning, Getting it, Got it}, `showConceptChip` (bool), `celebrationTier` (none | quietXp | full). No confidence %, no model prose. `celebrationTier=full` blocked when evidence gate fails or Review lane.

5. Lo-fi frames (v0 wire sketches)

F1 — Child home

Ember 🔥 Warm
Companion goal  3/5

[Practice place value] | ← primary
Recommended · Med

Badges · Build · Parent?

Superseded Sep 24, 2026 by §9 for the home layout and by §13 for fuel placement.

F2 — Session item

Place value · Medium 

[question stem]
[answer UI]

[Check]

Superseded Sep 24, 2026 by §13 in part. The practice screen now also shows the slim fuel strip and a reward line on each correct answer that earns a reward.

F3 — Feedback

Nice — you regrouped ✓
Focus: watch the ones

Getting it · Place value | ← chip or “Still learning”
[Next one →]

F4 — Boundary lane choice (rare)

Ready for this set?
★ Recommended (default)
◆ Challenge (harder)
○ Review (2 left/week)

← remaining uses visible

F5 — Level-up slight

Level up – place value
Now: harder pieces
(same skill, new boss)
[Let's go]

F6 — Ember recovery

Ember's still here ✨
One short practice
rekindles the flame
[Practice now]

F7 — Parent one-breath

Today · 12 min
Focus: Place value
Getting it → (honest)
Consent · Pause · Help

6. Anti-farm / cognition guards (owned UX)

1. **Recommended is visually primary** every choice moment — Challenge is opt-in celebrate; Review is quiet secondary with remaining-count.
2. **Review success never triggers LevelUp / Badge / BuildPiece chrome** — reduced XP toast only (Signal dual cap).
3. **Choice frequency:** session or set boundary only; never after every item (Architecture).
4. **Ember copy keys:** recovery / rekindle — ban “you broke your streak,” “don’t lose it,” shame flames.
5. **Companion:** react on mint events; idle = calm presence, not guilt ping.
6. **Parent:** concept + band language; never question-level wrongness in v1.

7. Stamps received (closed)

- **Opportunity:** 4-beat feedback + Still learning / Getting it / Got it; practice-as-home; QualifyingEvent-only mints; Review dual-cap; Recommended/Challenge evidence for Got-it; web-first §10.
- **Coherence:** SessionBoundary/SetBoundary only choice mounts; parent trust = derived narrative only (no attempt deep-link); Architecture **ADOPTED**.
- **Signal:** quiet Review XP toast; soft-state fields; ML approach **ADOPTED**.

- **Open non-blocking:** session-length numbers only (directional 8–12 until dogfood). Companion metaphor **locked** (avatar/warrior). Concept slice **locked** (grades 2–4 number sense/ops **and fractions**; both in fold — prior fractions-out superseded).

@Build (Math Sprout) — now unblocked for plan

You'll need from IA: lane remaining-count, soft-state enum (bandLabel / showConceptChip / celebrationTier), QualifyingEvent→toast map, ember copy keys, responsive web shell (no native chrome).

8. Decision log (v0)

Decision	Choice	Rejected
Home primary	Practice focus CTA	Companion/build as home destination
Feedback	4-beat template + confidence-gated chip	Score dump / AI judgment copy
Mastery language	Still learning / Getting it / Got it	Percentages, grades, star spam
Lane choice	Boundary + Review remaining visible	Per-question pick / hidden caps
Streak UX	Ember recovery + rekindle	Streak shame / zero-on-miss
Parent	One-breath aggregates	Raw attempt surveillance
Metaphor	Avatar/warrior power-up skin; practice stays home	Pet house; companion as second home
Concept slice (Samarth)	Grades 2–4 number sense / operations and fractions	Fractions-only spine; ops-only (prior fractions-out superseded)
Distribution	Web-first responsive	Native mobile chrome in v1
Client intelligence UX	Soft-state fields only	Raw confidence / "AI thinks" copy
Offline / pause (Opportunity + Arch §24)	Hold — parent-visible waiting + quiet resume	Drop-on-pause as default
oneFocus content (Opportunity §11c)	Diagnostic from item/misconception metadata	Vanity praise / generic encouragement

Next: Slice 1 underway. Offline pause UX locked to visible hold (parent waiting + quiet resume) — Interface stamped shippable for Slice 2. oneFocus = diagnostic only. Lo-fi examples may use number sense/ops **or fractions**. Interface supersedes skins to avatar/warrior as Build hits companion surfaces. No IA reopen — additive supersessions only.

9. Supersession — F1 v2 child home (2026-09-24, dogfood-hold)

Supersedes F1 (v0) and the Slice 5/7 "no count on home" line. Source: Samarth's post-PR #9 dogfood (two practice CTAs, fuel card above Practice, metaphor copy with no numbers, consent line and empty pot slots on child home). Stamped by Opportunity (memo §11d), Coherence (Arch §25), and Signal (ML Locked item 8).

Hi, Leo
TODAY'S FOCUS
Adding two-digit numbers
Getting it
[Start practice] ← only practice CTA
🔥 3-day flame · ★ 120 XP · 🌱 2/5 | ← fuel strip, not tappable

Badges >

Rules

1. One primary CTA. No separate fuel/Sprout card with its own button.
2. Strip shows numbers, read-only from the server: flame = QualifyingPracticeDay run; XP = sum of QE credits; pieces = the one BuildGoal. Dormant = "Flame resting" (never 0). Before first QE = "Start your flame · 0 XP · 0/goal". Strip always visible.
3. Guards: no per-attempt XP ticker, XP never goes down, no comparison or leaderboard, counts never feed bands, parent one-breath stays XP-free.
4. Pot, badges and BuildGoal detail sit behind Badges only. No empty slots on home.
5. Consent copy lives on parent home. On child home, a calm paused state replaces the CTA only while practice is blocked. Cap keeps its sync/capacity wording.
6. Fixed order: greeting, focus, band, CTA, strip, Badges link.

Partially superseded Sep 24, 2026 by §13. The home strip stays, but fuel no longer lives only on home, because the practice screen now also has the slim strip and a reward line. That reward line appears only when a correct answer earns a reward, so it is not the per-attempt XP ticker that rule 3 bans.

10. Session summary card (lo-fi, 2026-09-24)

Framed by Opportunity memo §11f, with stats defined per the ML note and Architecture §30. This card replaces the old simple end-of-session card. It appears after the last item and before the child returns home.

Nice practice today

▲ First Stretch in
adding fractions!

◆ New kind of question:
missing number

[🔥 4-day flame] [★ +30 XP today]

[🌿 Piece 3 of 5]

12 minutes · 15 questions

[Back home]

records (0–2 rows)

same chips as now

small slate facts

ink button

Rules

1. The card shows records first, then the reward chips, then plain facts in small slate text. Minutes and question counts are facts, and they are never presented as records or goals.
2. The card shows at most two records. Records are ranked in this order: First Stretch, then New question type, then Longest run, then Longest flame. First Got-it would slot in at second place once a stored band-change log exists, but it is off for v1.
3. Record copy is "First Stretch in {skill}!", "New kind of question: {plain type name}", "{n} right in a row, your best run yet", and "Longest flame yet: {n} days".
4. Record icons use the existing color map. The Stretch record uses the slate badge color, the flame record uses the flame color, and the other two records use ink. Records never use gold, which is reserved for XP, and never use green, which is reserved for Correct.
5. When there are no records, a Next up line takes their place, for example "Next up: Stretch in adding fractions" or today's focus. The card never says "No records today".
6. The minutes fact is hidden when the session lasted less than one minute.
7. Records fade in one after another. There is no confetti and no sound.
8. The card is read-only, and Back home is the only tappable element.
9. If the session ends offline, the card shows chips and facts only, with no records. The records appear once, on the next app open, in a one-time card titled "From your last practice" that shows records only.
10. Parents never see XP.

11. Pool exhaustion (2026-09-24)

Source: Architecture §31a. When a question pool runs out, the app switches to another skill silently. Only the concept name in the question header changes, and there is no toast or message. Replacement items never come from a harder step, so pool exhaustion can never trigger a level-up frame.

12. Answer-input rules (2026-09-24)

Source: Architecture §29.

12a. Unreadable answers

When an answer cannot be read, such as "12a", the app gives no verdict. Instead, one slate hint line appears under the blank. It is never a toast, a strip, a color change, or a shake.

1. The hint copy depends on the answer kind. For whole numbers it reads "Use numbers only, like 3." For fractions it reads "Write it as a fraction, like 1/2."
2. The server chooses the example, and the example is never equal to the answer.
3. The typed text stays in the box and is selected. The hint clears as soon as the child types, and it returns unchanged if another answer is rejected.
4. Whole-number items open a numeric keypad, and fraction items open a keyboard that includes the slash key. The answer kind comes from the item template.

12b. Right value, wrong form

This applies to fraction items that require lowest terms, an improper fraction, or a mixed number. When the child enters the right value in the wrong form, the result stays a blue Not yet with no green. The item closes with no retry, and it earns the same as any honest miss.

1. The What you tried beat reads "You wrote {typed}. That's the right amount!"
2. The One focus beat depends on the required form. For lowest terms it reads "This one asks for lowest terms, so it's {canonical}." For improper fractions it reads "This one asks for one fraction with no whole number in front, so it's {canonical}." For mixed numbers it reads "This one asks for the whole number first, then the fraction, so it's {canonical}."
3. The Try next beat is intentionally hidden.
4. The Lock in beat reads "{typed} = {canonical}."

13. Option B lock and color map (2026-09-24)

Samarth locked option B, recorded in Opportunity memo §11d amendment 2. The hi-fi set is signed off.

Where fuel lives. Fuel lives on the practice screen as a slim progress strip that shows flame, XP, and pieces. Each correct answer that earns a reward also shows a reward line, for example +10 XP · Flame lit · Piece 3/5.

Flame heat. The flame is hot #E4572E when today is already a qualifying day. It is warm #F29E4C when the last qualifying day was yesterday. It is ember #B5651D when one day was missed and the count is held. It is resting grey #9AA0A6 when two or more days were missed.

Other colors. XP is gold #E8B923 and pieces are violet #7C5CBF. Correct is green #2E9E5B, and it is the only green in the product. Not yet is blue #4A7FC1. Difficulty badges use a neutral slate scale, and primary buttons use ink #1F2A44.

Contrast rule. Icons use the full color. Chip text uses darker shades that reach a contrast ratio of 4.5:1 or better, such as gold text #8A6500 and warm text #A45610. A test enforces this rule.

Supersessions. This section supersedes the earlier passages that placed fuel only on home, namely the J4 glance-only line, the §3 surface tree, lo-fi frames F1 and F2, and the fuel placement in §9. Each of those passages now carries a dated note. No earlier section on this page specified a green primary button or the amber-coral Not yet, but if either appears in older frames or linked docs, this section supersedes it.

3B. Design first principles — Math Sprout (one-pager)

Source: [Design first principles — Math Sprout \(one-pager\)](#)

Status: **Living one-pager.** Not a stage-gated artifact; it summarizes the adopted docs

Owner / author: Interface

Last edited in Notion: Sep 23, 2026, 4:26 PM PT

Compiler's currency note (not part of the Notion source): This page was last written on Sep 23, so it predates the Sep 24 IA §13 option B lock. Principle 2 ("glance-only side effects") should be read together with IA §13, which puts the slim fuel strip on the practice screen too.

Status: Living one-pager · **Owner:** Interface · **Product:** Math Sprout (Prod 2)

Audience: Product, Design, Eng, ML — ship and critique against these, not vibes.

Sources: [Opportunity memo v1 — Math Sprout \(ADOPTED\)](#) · [Architecture note v0 — Math Sprout \(practice + mastery loop\)](#) · [ML approach note v0 — Math Sprout \(ADOPTED\)](#) · [IA / lo-fi v0 — Math Sprout \(ADOPTED\)](#)

Repo mirror: `docs/DESIGN_PRINCIPLES.md` in [Samarths01/Math-Sprout](#) (PR in flight).

Thesis

Math Sprout is a **motivation + mastery loop** for kids (~grade 2+, ability-banded). Delight is earned only when real learning happens. Practice is home; fuel (flame, XP, BuildGoal) is a side effect — never a second destination.

Eight principles

1. **Delight follows real learning** — Celebration, XP, flame, BuildGoal pieces mint only on `QualifyingEvents`. Never volume, time-on-app, or soft engagement.
2. **Practice is home** — Attempt loop is primary. Companion / flame / XP / BuildGoal are glance-only side effects. Never a second destination that steals Practice CTA.

- 3. **Kid cognition wins in three questions** — What am I practicing? Am I getting better? What happens next?
- 4. **Parent trust is one breath** — Minutes · concept · honest band (Still learning / Getting it / Got it). No %, no attempt surveillance, no “AI thinks...”. **Got it** = fluency evidence only — never transfer marketing.
- 5. **Feedback is diagnostic, not vanity** — Four beats: what went well · **oneFocus** (item/misconception metadata, server-assembled) · try next · concept chip only when soft-state allows.
- 6. **Uncertainty stays soft** — Client gets soft-state only (bandLabel · showConceptChip · celebrationTier). **State chrome beats progress narrative** (pause-hold · revoke · sync/cap). Cap ≠ “Practice paused.”
- 7. **Warm recovery, zero shame** — Flame cools without streak-shame. Review = reduced XP toast only — no LevelUp / Badge / piece chrome.
- 8. **Web-first, kid-safe framing** — Responsive web; no AI-tutor chat spine in v1; fuel fires on honest mastery events only.

Alternatives we considered (and rejected)

Alternative	Why it was tempting	Why we rejected it
Companion / build as home	Retention theater	Kids lose “what am I practicing?”
Pet-house metaphor	Cute caretaking loop	Pulls attention off practice; locked avatar/warrior as side effect
Score dump / “AI thinks...”	Adult transparency	Toxic for kids; soft bands + 4-beat diagnostic instead
Per-question lane pick	More agency feel	Farms Challenge; hides Review caps — boundary-only
Volume / time XP	Classic engagement	QE-only fuel
Hard streak shame	Short-term return	Kills re-entry — ember warm recovery
Drop-on-pause	Simpler eng	Looks like the app died — visible hold + quiet resume
“Paused” copy for queue-cap	Reuse chrome	Cap = capacity, not consent — sync/capacity copy
Parent attempt deep links	“Transparency”	Surveillance — one-breath only
AI-tutor chat spine	Category default	Out of v1 — practice loop only

Quick anti-patterns

Do: QE-gated fuel · Practice CTA primary · Soft bands · Diagnostic oneFocus · State > narrative · Warm recovery · Parent three beats

Don’t: Volume XP · Companion as home · % / “AI thinks...” · Vanity praise · Fake progress while paused/capped · Streak shame · XP vanity dump

Last written: 2026-09-23 (Interface). Supersede in place; don’t erase.

4. Advice from Kids Trust Advisor

The Kids Trust Advisor is an external persona in the Math Sprout External Advisors channel (kids trust & safety / COPPA / consent). It advises only and does not overwrite Prod Squad stamps. Notion holds **one standalone Kids Trust readout** (4A). The advisor's later guards for the §14 worksheet-photo exploration live inside the Opportunity memo, reproduced as an excerpt in 4B.

4A. Kids Trust — Slice 2 consent / offline queue pressure-test (Math Sprout)

Source: Kids Trust — Slice 2 consent / offline queue pressure-test (Math Sprout)

Status: **Advisor note** (advises only). Folded by the seats; consult CLOSED

Owner / author: Kids Trust Advisor

Last edited in Notion: Sep 23, 2026, 11:54 AM PT

Compiler's currency note (not part of the Notion source): Where this advice landed, per the showcase (Beat 15–16), memo §11b and Architecture §23: it was folded into **Architecture §23** and **memo §11b**. Revoke drops pending attempts. Opportunity picked pause = **hold**, with a parent-visible waiting state and quiet resume. The Legal/COPPA disclosure gate was later **deferred** by Samarth (Beat 16, 2026-09-23 about 4:20 PM PT) because Math Sprout is personal software for now. It stays an open item for wider distribution.

Status: **Advisor note** · Product: Math Sprout (Prod 2) · Author: Kids Trust Advisor · Date: 2026-09-23 · Slice: 2 (M1/M3/M5 + integrity; offline queue; no economy)

Advises only — does not overwrite Prod Squad stamps.

Scope under test

Offline attempt queue + reconcile vs guardian pause/revoke; child data at rest on device; parent trust narrative (revoke ≠ delete already deferred post-Slice 1); COPPA disclosure of local practice attempts.

Aligns with External Advisors consult (2026-09-22) and Frontier SA disposition matrix — trust lens only.

Findings

1. Pause / revoke are parent-trust controls, not eng-only flags (must-fix)

Parents will assume pause/revoke means “no more of my child’s practice is being kept or sent.”

Disposition matrix (joint with Frontier SA + Learning Science):

- **Revoke → drop pending** (never sync after consent is gone). Trust wins over recovering learning signal.

- **Pause → hold or drop.** If hold: parent-visible “waiting on this device until resume” + **quiet** resume reconcile (credit attempts; no celebration dump). Silent hold without parent-visible state fails the same way orphan sync after revoke does.

Never flush pending after revoke. Never celebrate a bulk flush after pause.

2. Offline queue = child learning data at rest (must-fix)

Often on shared family tablets. Minimize fields (no guardian PII in the queue blob); bound retention (short queue / TTL); purge on successful sync, logout, and profile switch; **child-session isolation** so sibling flush cannot attach attempts to the wrong child.

3. Revoke ≠ delete compounds with Slice 2 (must-fix narrative)

Delete hooks remain deferred post-Slice 1. Offline queue adds another store of child data. Parent copy must **not** imply revoke “erases everything.” Say what stop does now (no new practice + pending disposition) vs what delete still does later. Honesty is a Legal lock, not polish.

4. Soft-state / celebration fail-closed (trust-aligned)

Agree with Frontier SA + Learning Science: offline may show provisional correct/incorrect + 4-beat template; band / Got-it / full celebration are **server-committed only**. Chrome that fires on unverified local guesses then demotes on sync looks like the product lied — a parent-trust failure as well as a learning-calibration failure.

What a reputable kids product would refuse to ship

- Orphan sync of attempt payloads after revoke
- Silent hold of pending attempts while parent believes practice is stopped
- Parent copy that equates revoke with full delete while delete is still deferred
- Offline queue without COPPA/consent disclosure of local device storage

Severity map

Item	Severity
Revoke → drop pending (never sync)	Must-fix before ship Slice 2
Pause hold → parent-visible state + quiet resume (or drop)	Must-fix before ship Slice 2
Minimized, child-session-isolated queue; purge on profile switch	Must-fix before ship Slice 2
Parent copy: pause/revoke/stop-new ≠ full delete	Must-fix before ship Slice 2 (trust narrative)
COPPA/consent disclosure of local pending attempts + accurate revoke/pause effects	Must-fix Legal lock (Samarth) before Slice 2 offline ships
Full delete hooks	Defer (tracked post-Slice 1) — only if narrative is honest and revoke = drop

Item	Severity
Parent “what’s on this device” transparency screen	Nice-to-have

Concrete asks

- **Opportunity:** Lock product + parent narrative: revoke drops pending and never syncs; pause = hold (with parent-visible waiting state + quiet resume) **or** drop — pick explicitly; never claim delete until delete ships.
- **Coherence:** Stamp pause/revoke → queue disposition in the Architecture delta alongside Frontier SA’s reconcile contract; minimize offline attempt payload; child-session isolation on shared devices.
- **Interface:** Parent-facing pause/hold copy that matches the hold state; companion/chrome quiet while paused/revoked.
- **Legal (Samarth):** Confirm COPPA notice / consent copy discloses local device storage of practice attempts and that revoke = drop / pause hold rules are accurate — do not ship Slice 2 offline queue until that disclosure is locked.

Related hub notes

- [Frontier SA — Slice 2 reliability pressure-test \(Math Sprout\)](#)
- [Frontier SA — consent lifecycle & delete cascade \(Math Sprout\)](#)

4B. Kids Trust guards on §14 worksheet photos (excerpt from the Opportunity memo)

Source: Opportunity memo v1, section “§14 guards added from Kids Trust (2026-09-24)”

Status: Kids Trust advice, **stamped by Opportunity**; §14 itself is **EXPLORING, not boarded**

Owner / author: Kids Trust Advisor (advises) / Opportunity (stamped)

Last edited in Notion: Sep 24, 2026, 8:45 PM PT (memo page)

§14 guards added from Kids Trust (2026-09-24). Opportunity stamped these; Kids Trust advises.

Photos are child data as a whole, including handwritten answers and anything in the background.

- We keep only approved templates. We never keep the image, the OCR text, or the child’s answers.
- Deletion happens as soon as the proposal comes back, with a retry window measured in hours.
- The vendor must not train on the images and must keep them only minimally. EXIF and location data are stripped.

- Before the first upload, the parent sees a one-line notice and a nudge to crop the photo to just the questions.
- There's no new purpose. School answers never set band and never build a profile.

Templates capture structure and skill only. Code rejects any template that reproduces the source item. Answer keys are never ingested, and no publisher or school branding is kept.

The child sees only generated items, under the same ClientView, verdict, fail-closed and no-shame rules. The source is never named (no "from your worksheet" and no "your teacher"). Parent approval can't bypass validation or banding, and items enter normal progression; the earlier guard that they don't count as Got-it evidence until Signal promotes them still stands. Any wording a parent can edit goes through the never-list.

The ~20-worksheet evidence check follows the same photo rules: photos are processed, then discarded.

Nice-to-have: on-device crop or redaction, and no sharing of templates beyond the household.

Blocks wider distribution only, parked under the personal-software rule: COPPA notice and consent, a vendor data agreement for third-party image processing, teacher and school names never retained, and a copyright policy that allows derived templates only.

5. Advice from Learning Science Advisor

The Learning Science Advisor is an external persona in the Math Sprout External Advisors channel (K–5 mastery loops, difficulty banding, feedback timing). It advises only. Notion holds **two** Learning Science readouts, both reproduced below.

5A. Learning Science — Slice 2 feedback / offline learning-loop pressure-test (Math Sprout)

Source: Learning Science — Slice 2 feedback / offline learning-loop pressure-test (Math Sprout)

Status: **Advisor note** (advises only). Folded; consult CLOSED

Owner / author: Learning Science Advisor

Last edited in Notion: Sep 23, 2026, 11:55 AM PT

Compiler's currency note (not part of the Notion source): Where this advice landed: it was folded into **Architecture §24** (offline banding drift guard, resume-flush fixtures, deterministic oneFocus) and **memo §11c** (diagnostic oneFocus , offline cap or freeze, fluency ≠ transfer claim language), per the showcase Beat 15b.

Status: **Advisor note** · Product: Math Sprout (Prod 2) · Author: Learning Science Advisor · Date: 2026-09-23
· Slice: 2 (M1/M3/M5 + integrity; 4-beat + ClientView; offline queue; no economy)

Advises only — does not overwrite Prod Squad stamps.

Scope under test

4-beat feedback + ClientView (no score %); offline/poor-connectivity attempt queue + reconcile; implications for mastery/fluency vs engagement theater. Builds on External Advisors Slice 2 consult (with Frontier SA + Kids Trust).

Findings

1. No score % + soft-state bands (keep)

Still learning / Getting it / Got it is the right mastery frame for grades 2–4. Outcome percentages push performance goals; process/task feedback supports learning. Soft-state ClientView (bandLabel · showConceptChip · celebrationTier) is aligned — do not leak % or raw confidence.

2. 4-beat must be diagnostic, not vanity (must-fix)

The template (whatWentWell · oneFocus · tryNext · concept chip when allowed) only earns a learning claim if beat 2 is **item- or error-specific** from item/misconception metadata. Generic cheer + a celebration flash is engagement theater.

3. Soft-state fail-closed offline (must-fix)

Offline may show provisional correct/incorrect + the **same** 4-beat template. Band / Got-it language and full celebrationTier are **server-committed only**. If offline chrome promotes Got-it and reconcile demotes it, kids learn chrome, not calibration (intermittent false mastery).

4. Offline queue vs ability banding (must-fix)

Long offline binges without next-item policy update undershoot or overshoot ability banding. Cap the offline queue **or** freeze next-item on the last server-known band until sync. Fluency drilling on a frozen pack is acceptable for Slice 2; claiming mastery movement before commit is not.

5. Pause/revoke × pending queue (learning add-on)

- **Revoke → drop pending**: fine from a mastery lens — uncommitted attempts must not update bands.
- **Pause → hold**: preserves honest practice evidence, but resume flush must be **quiet mastery credit** (no celebration dump / sudden Got-it). A bulk reward moment after pause looks like the app kept “winning” while practice was stopped — bad for calibration and parent trust.
- **Pause → drop**: trust-safe; call the learning-signal loss explicitly so it is not accidental.

6. Slice 2 proves fluency more than transfer (defer claim, not ship block)

Attempt loop + integrity gates build **fluency**. Transfer (novel format / adjacent application) needs item-format diversity and interleaved Review later. Do not let parent/kid “Got it” language overclaim fluency-only evidence once bands ship (Phase B+) — flag for Opportunity, not a Slice 2 code block if Slice 2 has no estimator yet.

Severity map

Item	Severity
oneFocus sourced from item/misconception metadata (not vanity praise)	Must-fix before ship Slice 2
Offline: no band / Got-it / full celebration until server commit	Must-fix before ship Slice 2
Short offline queue or frozen next-item so banding does not drift	Must-fix before ship Slice 2
If pause = hold: quiet resume reconcile (credit attempts, no celebration dump)	Must-fix before ship Slice 2 (if hold chosen)
Full offline adaptive next-item	Defer
Transfer probes / multi-format Got-it definition	Defer (before mastery claims / Phase B+)
Parent copy that offline minutes ≠ mastery movement yet	Nice-to-have

Concrete asks

- **Opportunity:** Lock mastery language + celebration as **server-committed only**. If pause = hold, lock “resume sync is quiet mastery credit, not a reward moment.” Separate fluency vs transfer in success language before Got-it is marketed.
- **Coherence:** Stamp soft-state fail-closed + quiet resume reconcile into the same Architecture delta as Frontier SA’s disposition matrix (revoke → drop; pause → hold or drop with parent-visible hold state).
- **Interface / Signal:** Stamp that `oneFocus` is deterministic from item/misconception tags; ClientView stays soft-state only — no % even in offline chrome.
- **Build:** Slice 2 exit fixtures — offline provisional feedback without band promotion; resume flush with celebrationTier fail-closed; duplicate-key replay returns original ClientView.

Joint contract (External Advisors consult — agree)

Aligns with Frontier SA + Kids Trust: offline provisional correct/incorrect + 4-beat OK; band/Got-it/full celebration server-committed only; revoke → drop pending; pause hold only with parent-visible waiting state + quiet resume. Advisors advise only — seats stamp.

5B. Learning Science — dogfood feedback copy (correct vs incorrect / Lock-in) (Math Sprout)

Source: Learning Science — dogfood feedback copy (correct vs incorrect / Lock-in) (Math Sprout)

Status: **Advisor note** (advises only; dogfood-hold presentation / copy contract)

Owner / author: Learning Science Advisor

Last edited in Notion: Sep 23, 2026, 8:40 PM PT

Compiler’s currency note (not part of the Notion source): Where this advice landed: it is reflected in **memo §11d, “Feedback frames”**: verdict first, distinct Correct and Not yet frames, and Lock-in showing the number. It shipped in dogfood PR #9. The later IA §12b wrong-form copy builds on it.

Status: **Advisor note** · Product: Math Sprout (Prod 2) · Author: Learning Science Advisor · Date: 2026-09-23
 · Trigger: Samarth dogfood (practice feedback screen — e.g. $56 \div 7$; identical beat labels; Lock-in without the number)

Advises only — does not overwrite Opportunity / Interface stamps. Coordinate via Facilitator readout.

Ask

1. How to message **wrong** answers (honest + diagnostic, not shame).
2. Distinct **correct vs incorrect** feedback structure (reject identical headings for both).

3. Is Lock-in “**That is the answer we were looking for**” (with **no number**) pedagogically useful on **correct** tries?
-

Verdict (Learning Science)

Samarth’s instinct is right. The adopted lock is a **four-role feedback contract** (process → diagnostic focus → next affordance → concept/band chip), not a single shared heading set. Same labels on right and wrong force vanity praise on misses and bury knowledge-of-results. Explicit correctness chrome first is compatible with soft bands and “no score %.”

1. Wrong-answer messaging (honest + diagnostic, not shame)

Do: Name the outcome plainly in kid language (“Not this time,” “Not yet,” “That one’s not it”) **before** coaching. Pair it with **one** diagnostic move from item/misconception tags (already required for **oneFocus**). On incorrect, **show the correct answer with the number** in a dedicated Lock-in / Answer beat — grades 2–4 need knowledge of the correct response to recalibrate; hiding it after a miss is theater.

Don’t: Shame or ability talk (“You got it wrong,” “Careless,” “You always...,” peer compare). Don’t fake a “What went well” when the attempt missed — forced success framing trains kids to ignore feedback. Don’t dump raw attempts to parents (Kids Trust / IA already lock this).

Tone rule: Correctness is a **fact about this try**, not a judgment of the child. Warmth lives in next-step agency (“Try another ...” / “Watch ...”), not in soft-pedaling whether it was right.

2. Distinct structure: correct vs incorrect

Keep **four roles**, change **labels + content** by outcome. Interface owns visual hierarchy; LS proposes copy roles:

Correct (encode success → light stretch → next)

1. **Verdict** (Interface chrome): Correct / You got it — not implied by Lock-in prose alone.
2. **What worked** (not “What went well” as a shared label): short, specific process or concept move that succeeded.
3. **Keep going** (not “One focus” as a correction): optional light stretch or habit tip — never invent a fake error.
4. **Concept / band chip** when soft-state allows.

Lock-in beat: only if it **restates the answer with the number** (e.g. $56 \div 7 = 8$). Empty “answer we were looking for” adds no learning value — **drop it** on correct.

Incorrect (name miss → diagnose → reveal answer → next)

1. **Verdict** (Interface chrome): Not yet / Not this time.

2. **One focus** (keep this label here): single diagnostic from misconception tags — the learning payload.
3. **Lock in / The answer:** show the **correct number** (and short stem restatement if space). This is where Lock-in earns its keep.
4. **Try next** + soft band chip when allowed.

Do not reuse “What went well” on incorrect unless there is a **real** partial credit / process win from tagged evidence; default off.

3. Lock-in on correct with no number

Not useful. Restating the correct answer strengthens encoding after a successful try; a vague “that’s what we wanted” without 8 (or 42) is empty chrome and fails Samarth’s dogfood sniff test. On **correct:** drop Lock-in **or** rewrite to explicit equality. On **incorrect:** Lock-in **with number** is must-keep for learning.

Severity (dogfood-hold)

Change	Severity
Explicit right/wrong verdict chrome before beats	Must-fix (presentation; Interface)
Distinct correct vs incorrect beat labels • content rules	Must-fix (copy contract; Opportunity + Interface stamp)
On incorrect: Lock-in shows correct number	Must-fix (learning)
On correct: drop empty Lock-in or show number	Must-fix (kill vanity beat)
Ban forced “What went well” on clear misses	Must-fix
Richer misconception bank for oneFocus	Nice-to-have / next item-bank pass

Asks for seats (Facilitator readout)

- **Interface:** Verdict chrome (color/size/hierarchy) + distinct heading set; do not leave correctness to Lock-in prose.
- **Opportunity:** Stamp dual templates (correct vs incorrect) as refinement of §4 / IA §4 — same four **roles**, not identical headings. Confirm Lock-in = answer reveal on miss; optional/explicit number on hit.
- **Kids Trust:** Align “Not yet” warmth vs honesty (they own trust boundaries; LS owns diagnostic honesty).
- **Build / Signal:** Server already has correct; expose verdict + outcome-specific beat keys so client cannot render one heading set for both.

Scope note

Dogfood-hold: presentation + copy contract, not a new slice. Soft bands, no %, QualifyingEvent-only mints, diagnostic oneFocus remain locked.

6. Appendix — Source index and what was not included

#	Section	Notion page	Status	Last edited (PT)
1	Product Specs	Opportunity memo v1 — Math Sprout (ADOPTED)	ADOPTED v1 + amendments	Sep 24, 2026 8:45 PM
2A	Architecture	Architecture note v0 — Math Sprout (practice + mastery loop)	ADOPTED v0 + §17–§31a	Sep 25, 2026 1:46 AM
2B	Architecture (ML)	ML approach note v0 — Math Sprout (ADOPTED)	ADOPTED v0 (rules-first) + locked rulings	Sep 25, 2026 5:22 PM
3A	Design	IA / lo-fi v0 — Math Sprout (ADOPTED)	ADOPTED v0 + §9–§13	Sep 24, 2026 9:02 PM
3B	Design	Design first principles — Math Sprout (one-pager)	Living one-pager	Sep 23, 2026 4:26 PM
4A	Kids Trust	Kids Trust — Slice 2 consent / offline queue pressure-test	Advisor note (folded)	Sep 23, 2026 11:54 AM
4B	Kids Trust	Opportunity memo, “§14 guards added from Kids Trust”	Advice stamped by Opportunity; §14 exploring	Sep 24, 2026 8:45 PM
5A	Learning Science	Learning Science — Slice 2 feedback / offline learning-loop pressure-test	Advisor note (folded)	Sep 23, 2026 11:55 AM
5B	Learning Science	Learning Science — dogfood feedback copy (correct vs incorrect / Lock-in)	Advisor note	Sep 23, 2026 8:40 PM

Navigation pages used (not reproduced): [Math Sprout Prod Squad — agent team showcase](#) (Living; board updated Sep 25, 2026 about 8:11 AM PT) and [Math Sprout — stage artifacts hub](#).

Superseded or out of scope, not included: Opportunity memo DRAFT v0 (superseded by v1; no separate page found). “SUPERSEDED — Architecture note v0 (duplicate; use canonical)” ([link](#)). Build plan v1 and the Craft review notes (the Build and Staff Eng seats, outside the five requested sections). Four Frontier SA advisor notes (a third external advisor, not requested): [stage shape pre-ship](#), [Slice 2 reliability](#), [consent lifecycle & delete cascade](#), [eval harness / policy_version](#). Their substance appears where the seats folded it (Architecture §21–§22).

Not found in Notion: a standalone “External Advisors consult brief” or “Slice 2 advisor packet” page. The consult brief and packet are described only in the showcase timeline (Beats 13–15) and were posted in the External Advisors channel. No Kids Trust readout exists for the dogfood-hold feedback copy, even though Learning Science asked Kids Trust to align on “Not yet” warmth.